

Hip pain

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...Lionel Messi returned to Barcelona early from international break due to an inflammation in his hip, pain that has also plagued other players such as Raul, Xabi Alonso, Aduriz, Busquets and a host of Athletic Club's current squad. There are different options in terms of treatment, and each player has undergone their own decision towards recovery to get back onto the pitch as quickly as possible.

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Recently, the hip joint has been recognised as a possible cause of problems in the athletic population. This is reflected in the published literature describing hip pathology and its treatment. The advent of magnetic resonance imaging (MRI) and then hip arthroscopy increased the awareness of hip labral pathology and anatomical variants that can lead to femoroacetabular impingement (FAI) syndrome as a potential underlying cause of hip and groin pain in young and middle-aged people. Lateral hip pain is also increasingly recognised as a substantial clinical problem.

EPIDEMIOLOGY

Pain reported by patients as being in the hip region ('hip pain' for convenience in this chapter) is a common cause of activity restriction in athletes. Hip and groin pain is the third most common injury reported in the Australian Football League (AFL),¹ accounting for between 5-15% of all AFL-related injuries,² and is commonly seen in many other sports, including tennis, football of all codes and hockey. People with 'cam'-type FAI are four times more likely to develop hip pain over 4 years than those without this morphological variant.³ Burnett et al. demonstrated that 92% of patients with arthroscopically confirmed labral tears complained of moderate to severe groin pain.⁴ Philippon et al. described labral tears and FAI in 100% of professional National Hockey League ice hockey players presenting for hip arthroscopy for the treatment of longstanding hip and groin pain.⁵ However, cam-type morphology is also associated with labral and cartilage hip damage in asymptomatic males where adjusted odds ratios range from 2.45 to 2.77.⁶ Therefore the relationship

between activity, pain, morphology and pathology is complex and not fully understood.

The likelihood of an athlete sustaining an injury to the hip joint can be increased by the demands of the sport, in particular sports that require repetitive hip flexion, adduction and rotation,⁷ but may also be caused by inherent or acquired individual anatomical variations within the joint, such as FAI or developmental dysplasia of the hip.^{5, 8-13}

Intra-articular hip pathologies, including chondral and labral pathology contribute to a reduced ability to participate in sporting or physical activities as well as pain and reduced function during activities of daily living.¹⁴ In addition, people with pathology such as FAI, labral tears and chondral pathology have impairments in hip range of motion, hip muscle strength, balance and functional task performance.^{14, 15} These physical impairments are themselves associated with further pain, limitations in daily activities, sporting participation and quality of life in affected individuals.¹⁶ Given the increased recognition of hip pathology as a source of hip and groin pain in active individuals, it is important that clinicians have a thorough understanding of likely diagnoses and appropriate treatment options in these patients.

FUNCTIONAL ANATOMY AND BIOMECHANICS

The hip has three functions, which are to allow mobility of the lower limb, to transmit loads between the upper body, trunk and lower limb, and to provide a stable base in weight-bearing activities. The anatomy of the hip allows it to perform these functions.

The hip joint is supported by a number of dynamic and passive supports, which include its bony morphology, passive restraints such as capsule and ligaments, and a complex system of interplaying muscle groups. The biomechanics of the hip joint are generally under-reported in the literature and as such are poorly understood. An appreciation of the functional anatomy of the hip and the role of the various structures surrounding the hip will assist in this understanding (Fig. 31.1).

Morphology

The hip joint (femoroacetabular joint) is a tri-planar synovial joint, formed by the head of femur inferiorly and the acetabulum superiorly. The acetabulum sits within the bony pelvis and is normally anteverted (forward facing) by approximately 23° (Fig. 31.2a).¹⁷ The acetabulum also faces inferiorly and laterally.

The head of femur is also anteverted, which refers to the angle between the axis of the femoral neck and the transcondylar axis of the knee (Fig. 31.2b). This angle is normally between $10\text{--}15^\circ$ in adults. The head of femur also faces superiorly and medially.

Any orientational or morphological deviation from this normal anatomy can result in unnecessary stresses within the hip joint. An undercoverage of the femoral head by the acetabulum as seen in acetabular dysplasia can place high static loads on the anterosuperior acetabular cartilage due to a decreased contact area. In contrast, an overcoverage of the femoral head by the acetabulum can lead to 'pincer'-type FAI. A cam deformity can lead to a biomechanically imperfect hip joint, as it may cause cam-type FAI. The anteversion of the proximal femur and acetabulum and the neck-shaft angle can in turn influence the susceptibility of the hip joint to FAI. The concept of FAI will be discussed in more detail below.

This morphological structure allows the hip to achieve its 3° of movement, flexion and extension, adduction and abduction, and external and internal rotation. It also allows the athlete to move freely in three planes of movement.

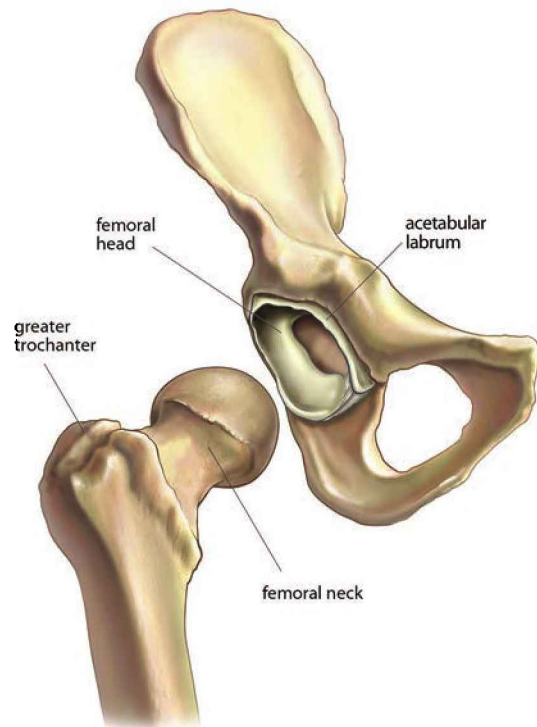
Acetabular labrum

The acetabulum forms the socket of the hip joint and is lined with articular cartilage. The acetabular labrum (Fig. 31.3) is a ring of fibrocartilage and dense connective tissue which is attached to the bony acetabular rim. The acetabular labrum is thinnest in its anterior aspect.

The blood supply of the labrum enters through the adjacent joint capsule. Only the outer third of the labrum is vascularised.¹⁹ The labrum is innervated by a branch of the nerve to quadratus femoris and also by the obturator nerve.²⁰ Nocioceptive free nerve endings are distributed throughout the acetabular labrum, while labral cells are capable of releasing cytokines when damaged, suggesting



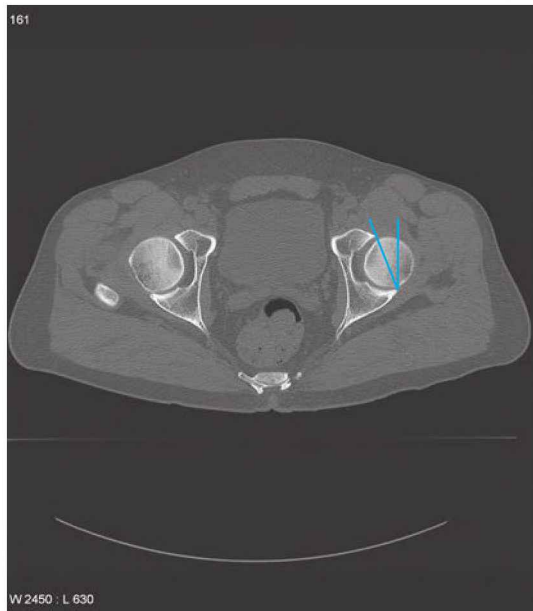
(a)



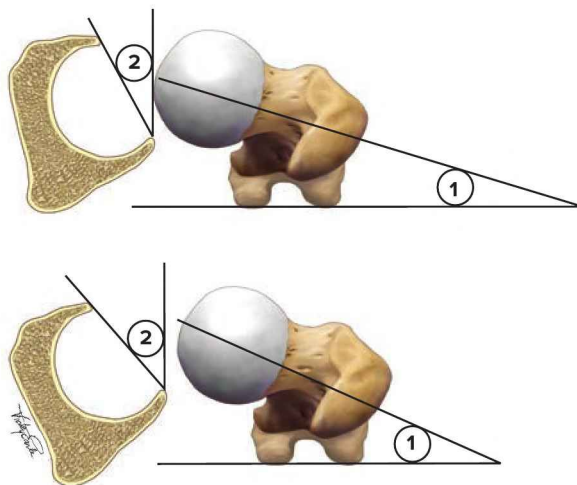
(b)

Figure 31.1 Anatomy of the hip and groin area (a) Plain radiograph of the pelvis (b) The hip joint

a pain and inflammation-producing capacity.²¹⁻²³ In addition, the labrum contains nerve end organs which suggest that, as well as a capacity to mediate pain, it can also assist in proprioception and generate neurosecretion that may assist in connective tissue repair.²⁰



(a)



(b)

Figure 31.2 (a) Computed tomography showing acetabular anteversion¹⁷ (b) Transverse views of a normal hip (upper figure) and a dysplastic hip (lower figure)¹⁸ (1) Angle of torsion—rotation of the femoral neck relative to the shaft (transcondylar axis), normally 10–15° of anteversion. Dysplastic hips usually have increased angle (2) Acetabulum anteversion angle—the anterior direction of the acetabulum is normally 20–40° of anteversion. Dysplastic hips usually have increased angle but it may be decreased

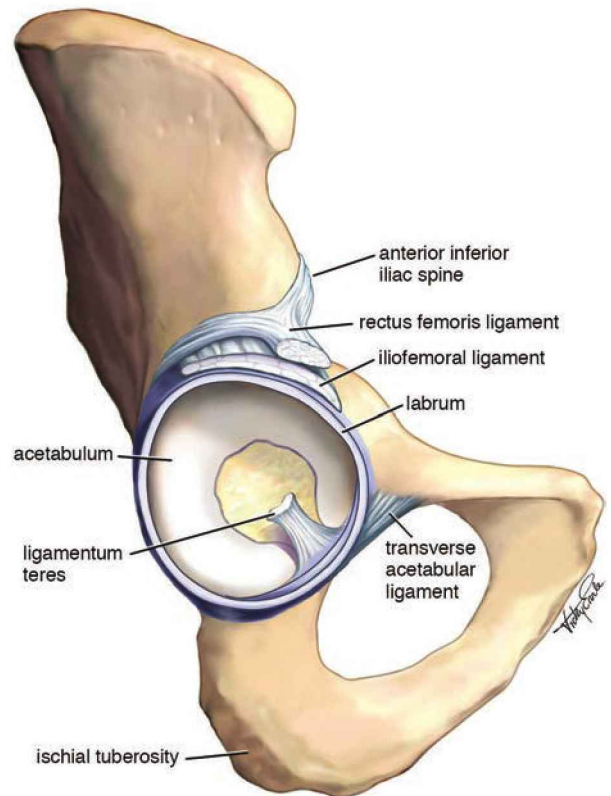


Figure 31.3 Transverse acetabular ligament, acetabular labrum and ligamentum teres (resected)

The acetabular labrum has several functions. These are primarily to deepen the acetabulum, to distribute the contact stress of the acetabulum over a wider area (increasing contact area by 28%)²⁴, and to assist in synovial fluid containment and distribution by virtue of the fluid seal mechanism.^{22, 24, 25} The acetabular labrum is continuous with the acetabular rim and adjacent acetabular articular cartilage, providing nutrition to, and assisting in, modulating loads within the acetabulum of the hip.

Normally, the fluid seal of the labrum and synovial fluid pressure is greatest in activity involving external rotation of the femur, while activity involving flexion and internal rotation (impingement) reduce the fluid seal and synovial fluid pressure.²⁶ Also people with cam-type FAI do not achieve maximal fluid seal and synovial fluid pressure during external rotation.²⁷ Given that adequate labral seal and synovial fluid pressure are essential for labral and cartilage nutrition, this has the potential to cause problems. These findings may help explain why hip impingement is associated with hip pain and pathology such as labral and chondral lesions, and why people with cam-type FAI are more susceptible to these associated pathologies.

Ligaments of the hip

The transverse acetabular ligament (Fig. 31.3) traverses the acetabular notch, connecting the anterior and posterior edges of the labrum. The deepest layer of labral tissue blends into this ligament. The transverse acetabular ligament is under greatest load in weight-bearing, when the head of femur relocates in the acetabulum, widening the acetabular notch and placing the transverse acetabular ligament under a tensile load.²⁵

Originally thought to be a histological vestige that becomes redundant early in childhood, it is now assumed that the ligamentum teres (Fig. 31.3) plays an important proprioceptive role, especially in weight-bearing activities.²⁸

The hip capsule contains ligaments that act together to restrain excessive motion of the hip joint. These are the iliofemoral ligament, the pubofemoral ligament and the ischiofemoral ligament (Fig. 31.4). The iliofemoral ligament ('Y ligament of Bigelow') reinforces the anterior capsule and originates from the anterior iliac spine, fanning into an inverted Y shape to insert into the intertrochanteric line. It is taut in hyperextension and also provides stability in relaxed standing. The pubofemoral ligament arises from the anterior surface of the pubic ramus and inserts into the intertrochanteric fossa. It is

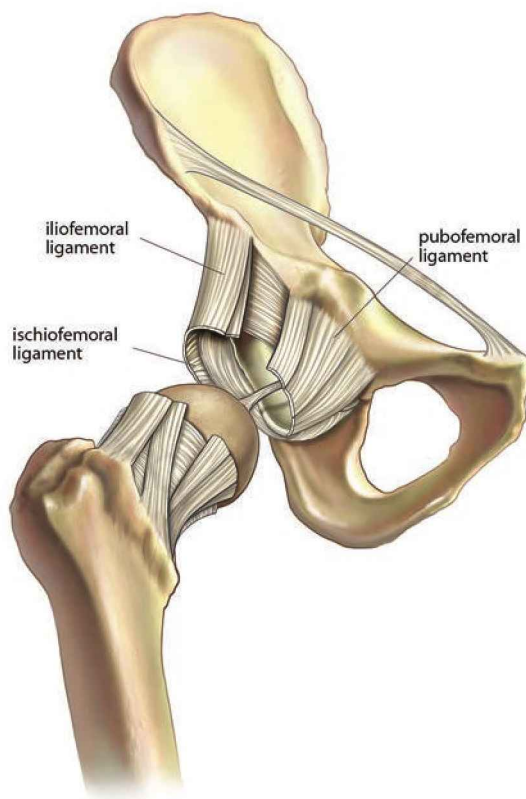


Figure 31.4 Capsular ligaments of the hip

taut in abduction and extension and also reinforces the anterior capsule. The ischiofemoral ligament arises from the posterior surface of the acetabular rim and labrum and extends into the femoral neck just proximal to the greater trochanter. Its fibres run in a spiral pattern and are also taut in hyperextension. These three ligaments act to restrain hyperextension, which is of particular relevance in relaxed standing.

Chondral surfaces

Both articular surfaces of the hip are lined with articular cartilage. These chondral surfaces rely upon adequate function of the synovium and movement of synovial fluid within the joint to provide nutrition, as articular cartilage is avascular. As detailed above, the degree of synovial fluid distribution is affected by the functional position of the hip, as well as by the presence of cam-type FAI, indicating that both the position of the hip and cam-type FAI may impact on cartilage nutrition. Both acetabular labrum and ligamentum teres have been reported to attach to synovium, and thus both may also play a role in the nutrition and normal function of articular cartilage within the hip joint.

In a normal, healthy hip joint, the articular cartilage assists in smooth movement as well as absorbing loads, protecting the subchondral bone and adjacent intra- and extra-articular structures. Glycosaminoglycans (GAGs) are molecules contained within the extracellular matrix of cartilage. They provide the strength and shock-absorbency characteristics typical of healthy cartilage. When reduced, the strength of cartilage and its ability to absorb forces are greatly inhibited. Lower levels of GAGs are reported in the early stages of osteoarthritis (OA) and also in people with cam deformity, pain on impingement testing and painful FAI.²⁹ There is an increased risk of chondral injury when coexisting labral pathology and cam-type FAI are present,¹⁴ while it has been suggested that chondral injury may represent hip OA in its earlier stages.³⁰

It appears that the interrelated nature of cam-type FAI, labral pathology and chondral injury all contribute to the breakdown of normal hip function and that abnormal hip mechanics may also contribute to these pathologies. Strategies that are focused on optimising the mechanics of the hip joint may be important targets when improving function, reducing pain and slowing hip-joint disease progression.

Muscle function

Dynamic hip stability is provided by a complex interplay between various muscles surrounding the hip joint. In particular, the primary hip stabilisers provide a posterior, medial and inferior force on the femur to control the

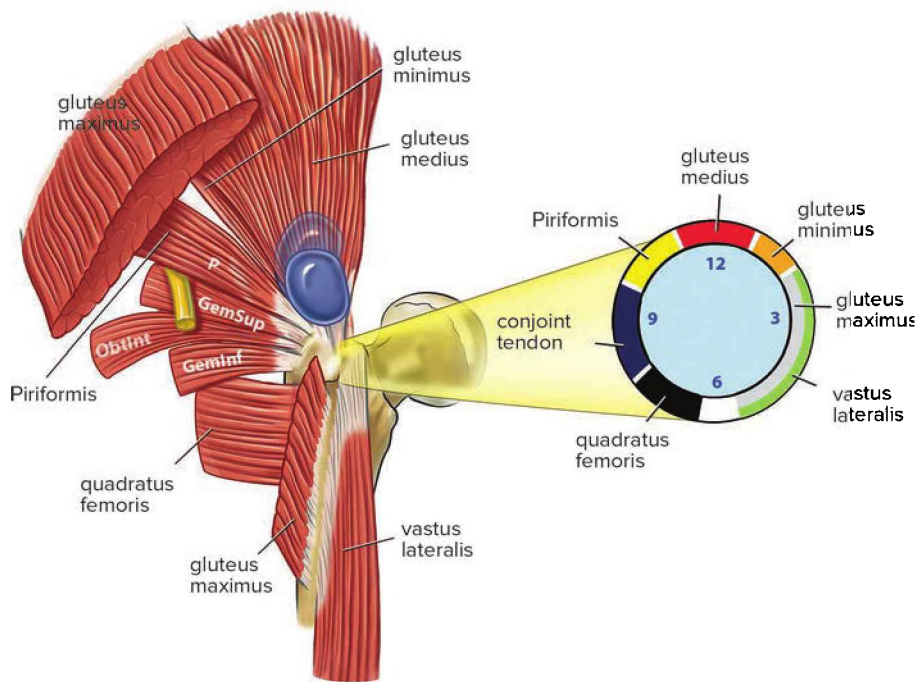


Figure 31.5 Muscle attachments around the greater trochanter

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The concept of deep hip stabilisers as the hip 'rotator cuff'³¹ has grown in popularity in recent years.

position of the head of femur within the acetabulum.³² Ultimately, the dynamic control provided by the deep hip stabilisers has potential to minimise stress on vulnerable structures, such as the anterosuperior acetabular labrum and the anterosuperior acetabular rim (Fig. 31.5).

Reports detail the role of hip muscles with respect to muscle morphology, primary action of joint movement and lines of action in relation to the axes of joint movement.^{31, 33, 34} Some muscles have greater capacity to generate torque over larger ranges of motion (prime movers), while other muscles are better placed to act as dynamic hip joint stabilisers.

Short hip stabilising muscles

The six short hip external rotators (obturator internus and externus, the superior and inferior gemellus, quadratus femoris and piriformis), also known as short hip stabilising (SHS) muscles, provide hip joint compression and hence dynamic stability during most weight-bearing and nonweight-bearing activities.³³ The gluteus medius is the dominant hip abductor and is the primary lateral stabiliser of the hip during one-leg stance activities.³⁴

Clinical biomechanics

For the patient with hip pain and/or pathology, the clinician should also consider the lines of actions for each of the deep hip stabilisers.³³ For example, although all of the SHS muscles provide dynamic hip stability in the anatomic position, the quadratus femoris also has a line of action that is inferomedial. Therefore, it has a greater capacity to resist superior translation of the hip. Similarly, the gluteus maximus and four SHS muscles (piriformis, gemellus inferior and superior, obturator internus) have a line of action that is posteromedial and may be able to resist anterior force of the hip.³⁵

In contrast, although the gluteus medius is an important lateral stabiliser of the hip, its line of action is both medial and superior³³ and it is the greatest contributor to both medial and superior hip contact force during walking.³⁵ Furthermore, the anterior fibres of gluteus medius and minimus become hip internal rotators when the hip is flexed. The relevance of these factors to the rehabilitation of the patient with hip pain and pathology will be described below.

In healthy people without hip pain, men are generally stronger than women in most hip muscle groups, while there is no difference between the dominant and non-dominant leg.³⁶ Hip muscle strength can be measured reliably in the clinic using hand-held dynamometry.^{36, 37} In certain populations, the relationships between agonist and antagonist muscle groups have been reported,

providing further evidence of how hip muscles function together in healthy groups^{37, 38} and how this function might be impaired in people with hip pain and pathology.

CLINICAL APPROACH

Hip region pain can be multifactorial and therefore difficult to diagnose. Hip pain also commonly coexists with groin-related pathology, such as the adductor, iliopsoas, inguinal and pubic-related entities (Chapter 32). These entities can present as hip pain. This makes definitive diagnosis and provision of appropriate management programs difficult and often multifactorial.³⁹ Therefore, the key to the effective management of hip pain is an accurate diagnosis of the likely source of and reason for the pain and understanding important differential diagnoses. The causes of hip pain are shown in Table 31.1.

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One major issue facing clinicians when encountering a patient with pain in the hip and groin region is the poor specificity of the majority of clinical tests,⁴⁰ making accurate diagnosis of the source of pain difficult (see also Chapter 14).

Fortunately, a number of clinical tests have good or excellent sensitivity, meaning clinicians can have a reasonable degree of certainty that a patient with a negative test finding does not have pain whose source is the hip joint.

A second issue is the potential for intra- and extra-articular sources of hip pain, as well as remote factors to contribute to the patient's symptoms. Finally, imaging modalities are useful, but are not equal or superior to arthroscopy when diagnosing hip pain. Therefore, clinicians need to consider a range of possible sources of pain when making the diagnosis from a combination of subjective and objective findings and imaging results.

Hip region pain and associated pathologies have not been well managed in the athletic population until recently. A large cohort study of patients with FAI in the USA indicated that most patients had hip pain for 12–36 months.⁴¹ Weir and colleagues reported a mean duration of hip and groin pain in athletes of 22 weeks,⁴² with the maximum duration 250 weeks (5 years), while many other studies report symptom duration of greater than 2 years.¹⁸ Byrd and Jones reported an average of 7 months from initial assessment and multiple other diagnoses being made before a definitive diagnosis of hip pathology was made.⁴³

In the fourth edition of *Clinical Sports Medicine*, we mentioned the debate regarding the efficacy of hip

arthroscopy as an appropriate intervention for athletes with hip pain. The efficacy of both surgical and non-surgical interventions for hip pain remains unknown. The danger of both surgical and non-surgical procedures being perceived as being fashionable is that practitioners can become polarised in their opinions and not assess each case and therefore the most appropriate intervention on its merits. Future studies determining the efficacy of both surgical and non-surgical management of hip pain are essential.

When assessing the hip, the reliability and clinical utility of assessment findings should be considered. These are contained in Table 31.2. A comprehensive discussion of the key components of a clinical history, physical examination and appropriate referral for investigations is outlined below.

History

The only feature from a clinical history with known validity is painful clicking; it has a low likelihood ratio (Table 31.2). This indicates that a patient without painful clicking is unlikely to have a labral tear. Nonetheless, it is important to obtain a full history from a patient prior to undertaking a physical examination or obtaining any investigations. This history should include:

- age, general health, past medical history (including presence or absence of childhood conditions such as 'clicky hips', slipped upper femoral epiphysis or infantile dysplasia), past family history of hip pain and hip OA (there is a greater risk of FAI if it is present in other family members⁵⁰) and medications
- weight and height (BMI >25 can increase severity of symptoms of OA and tendinopathy, as well as increase joint loads; also, patients with FAI and related pathology tend to have a BMI >25^{14, 41})
- exact mechanism of injury (if known), was onset sudden or insidious?
- time since onset of symptoms (patients with FAI often have had symptoms for 1–3 years⁴¹)
- pattern of symptoms since onset (worsening, improving or not changing)
- presence of mechanical symptoms such as locking, clicking or giving way (suspect labral or possibly ligamentum teres pathology)
- location of pain (hip pathology may present as groin, lower back, lateral hip, buttock or thigh pain⁵¹ and secondary sources of pain such as muscle spasm, which complicate the assessment)
- Note that lateral hip pain is usually *not caused* by hip joint pathologies

Table 31.1 Causes of hip pain

Common	Less common	Not to be missed
Anterior pain		
Labral tear Chondropathy Osteoarthritis Synovitis Ligament teres tear Hip joint instability (hypermobility or developmental dysplasia of the hip) Groin-related pain (see Chapter 32)	Stress fracture of the neck of femur (see Chapter 32) Traction apophysitis (anterior inferior iliac spine–rectus femoris; anterior superior iliac spine–sartorius; lesser trochanter–iliopsoas)	Avascular necrosis of the head of femur Synovial chondromatosis Slipped capital/upper femoral epiphysis Perthes' disease (see Chapter 44) Tumour
Lateral pain		
Gluteus medius tears and tendinopathy	Referred pain lumbar spine Pain from the trochanteric bursa	Fracture of neck of femur Nerve root compression Tumour
Posterior pain		
Posterior labral tear Posterior chondral lesion Ligamentum teres tear Posterior micro-instability secondary to anterior impingement	Referred pain lumbar spine and pelvic joints Proximal hamstring-related pain (see Chapter 34) Sciatic nerve entrapment in short posterior rotators (see Chapter 30)	Nerve root compression Tumour

- nature of pain (intensity, severity, constancy, time of day, latency will provide clues as to presence of inflammation, synovitis, bursitis or tendinopathy in addition to intra-articular pathology)
- neurological signs and low back pain (the lumbar spine can refer pain to the hip and should be eliminated as a potential primary source of pain)
- aggravating factors (be specific regarding position of hip and potential for impingement during these activities, how long it takes for these activities to provoke pain and latent pain)
- current level of activity (frequency and intensity, differentiate between upper and lower limb workouts. Tendon pain tends to be load dependent over a period of 3 days, thus aggravating factors related to gluteal tendinopathy may be difficult to determine if the person has very limited 'unloaded leg days')
- factors easing pain (be specific regarding positions of ease as well as time required for pain to ease)
- current sporting history (including level of sport, such as community, state, national; and position played

within the team as certain positions will place the hip under more load

- previous sporting history (certain sports played may increase the likelihood of a hip injury such as dancing, gymnastics, martial arts, tennis and hockey)
- desired level of future sporting activity (this is important to establish in order to determine level of intervention as well as future risk of injury)
- activities of daily living, including occupation, length of time spent sitting, amount of hip flexion and rotation, and degree of manual labour within occupation, family situation including the presence of young children
- any past treatment, including investigations, conservative treatment or surgical intervention.

During the history, appropriate patient reported outcome measures (PROMs) should be given to the patient to complete to establish a baseline score for severity of the condition, to explore specific areas of concern and to allow accurate monitoring of the patient's progress over time (see below).

Table 31.2 Diagnostic accuracy of clinical tests (history, physical exam) in patients with hip pain

Assessment finding	What is a positive finding?	Sn	Sp	LR–	LR+
History					
Labral tear					
Painful clicking in hip ⁴⁴	Clicking	100	85	0.00	6.7
Physical examination tests					
Intra-articular pathology (not specific)					
Flexion, abduction and external rotation (FABER) ⁴⁰	Pain	42–81	18–75	0.72–0.73	1.1–2.2
Scour test ⁴⁵	Pain	50	29	0.70	1.72
Thomas test ⁴⁶	Pain and reduced range	89	92	0.12	11.1
Labral tear					
Flexion, adduction and internal rotation (FADIR) (vs magnetic resonance arthrogram) ⁴⁰	Pain	94	9	0.48	1.02
FADIR (vs arthroscopy) ⁴⁰	Pain	99	5	0.15	1.06
Flexion internal rotation ⁴⁰	Pain	96	25	0.27	1.12
Femoral fracture					
Patellar-pubic percussion ⁴⁰	Reduced percussion on side of pain	95	86	0.07	6.11
Gluteal tendinopathy					
Trendelenburg ⁴⁰	Drop in NWB pelvis	61	92	0.25	6.83
Resisted hip abduction ⁴⁰	Weakness and pain	71	84	0.37	5.50
Resisted internal rotation ⁴⁷	Weakness and pain	55	69	0.66	1.77
Resisted external de-rotation test ⁴⁸	Pain	88	97.3	0.12	32.6
Single-leg stance hold for 30 seconds ⁴⁸	Pain during single-leg stance	100	97.3	0.0	37
Combination of findings					
Hip OA (vs X-ray)⁴⁹					
Squatting aggravates symptoms	5/5	14	98	0.87	7.3
Lateral pain on active hip flexion	≥4/5	48	98	0.53	24.3
Scour test positive	≥3/5	71	86	0.33	5.2
Pain with active hip extension	≥2/5	81	61	0.31	2.1
Passive internal rotation ≤25°	≥1/5	95	18	0.27	1.2

Sn = sensitivity; Sp = specificity; LR– = negative likelihood ratio; LR+ = positive likelihood ratio. High sensitivity indicates a test is useful for ruling out a condition (good screening tool), while high specificity indicates a test is useful for ruling in a condition (good diagnostic tool).

Blue = large change in likelihood (LR– <0.1, LR+ >10). This indicates that a positive test is likely to indicate a 'true-positive' finding, while a negative test is likely to indicate a 'true-negative' finding, meaning the test has high clinical utility.

Red = rarely important change in likelihood (LR– >0.5, LR+ <2.0). This indicates that a positive test is unlikely to indicate a 'true-positive' finding, while a negative test is unlikely to indicate a 'true-negative' finding, meaning the test has poor clinical utility.

Physical examination

In this section, the clinical tests that are valid are identified with a double asterisk (**). Other tests that have not had their validity reported but are likely to be useful are also listed.

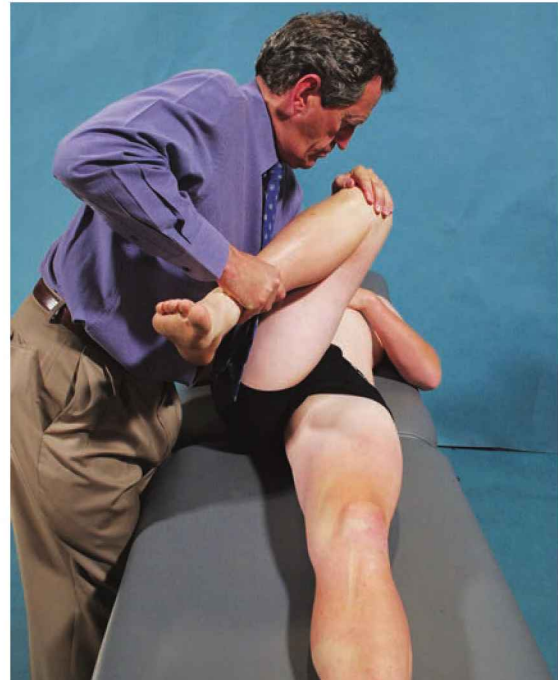
Examination involves:

1. Observation
 - a. standing:
 - i. general lower limb alignment
 - ii. femoral alignment
 - iii. muscle tone and symmetry
 - b. walking
 - i. pain
 - ii. limp
 - iii. lateral pelvic stability 'Trendelenburg sign'
2. Active movements
 - a. hip flexion/extension
 - b. hip abduction/adduction
 - c. hip internal**/external rotation at 90° flexion
3. Passive movements
 - a. flexion (Fig. 31.6a)
 - b. anterior impingement test**—flexion, adduction and internal rotation (FADIR) (Fig. 31.6b)
 - c. internal rotation at 90° flexion**
 - d. flexion, abduction and external rotation** (FABER or Patrick's test) (Fig. 31.6c)
 - e. quadriceps muscle stretch
 - f. psoas muscle stretch/impingement (Thomas position) (Fig. 31.6d)
4. Tests of muscle function
 - a. adductor squeeze test** (Fig. 31.6e)
 - b. hand-held dynamometry of hip muscle strength bilaterally
 - i. flexion/extension
 - ii. adduction/abduction
 - iii. internal/external rotation



(a)

Figure 31.6 (a) Measurement of hip flexion range of motion



(b)



(c)

Figure 31.6 (cont.) (b) Passive movement—anterior impingement (hip quadrant: flexion, adduction and internal rotation—FADIR) (c) Passive movement—flexion, abduction and external rotation (FABER or Patrick's test). Range of motion, apart from extreme stiffness/laxity, is not that relevant. Some caution needs to be exercised, as it is possible to sublux an unstable hip in this position. Pain felt in the groin is very non-specific. Pain in the buttock is more likely to be due to sacroiliac joint problems. However, pain felt over the greater trochanter suggests hip joint pathology



(d)



(e)

Figure 31.6 (cont.) (d) Passive movement—psoas stretch (Thomas position). Pain in the hip being stretched suggests psoas abnormality. Pain in the hip being compressed can be significant for anterior impingement of the hip joint (e) Resisted movement—squeeze test. Examiner places fist between knees as shown. Patient then adducts bilaterally against the fist (f) De-rotation test—in 90° flexion, the hip is taken into external rotation and the patient asked to return the leg to the axis of the table against resistance. The test result is positive when the usual pain is reproduced (g) Palpation of iliopsoas in muscle belly and at anterior hip joint



(f)



(g)

- iv. de-rotation test⁴⁸ (Fig. 31.6f). In 90° flexion, the hip is taken into external rotation and the patient asked to return the leg to the axis of the table against resistance. The test result is positive when the usual pain is reproduced.
5. Palpation
 - a. anterior hip joint line
 - b. adductor muscles/tendons/entheses
 - c. pelvis including pubis symphysis, ischial tuberosities and proximal hamstring attachment
 - d. iliopsoas in muscle belly and at anterior hip joint (Fig. 31.6g)
 - e. superficial hip abductors including tensor fasciae latae, gluteus medius and superior gluteus maximus
 - f. greater trochanter and tendons of gluteus medius and minimus
6. Functional movements
 - a. double-leg and single-leg squat, looking at range of motion, femoral alignment and balance
 - b. hopping (to reproduce pain)
 - c. forward hop
 - d. step up and down forwards on the affected leg (observe stability, pain level and pain location)
 - e. side step up and down on the affected leg
 - f. hip hitch (in neutral, internal rotation), keeping the knee extended. Identify the pain location and severity. Reproduction of sacroiliac joint or low back pain probably indicates a low back or sacroiliac joint problem rather than a hip problem
 - g. kicking (if appropriate)
 - h. balance and proprioception in dynamic tasks

- i. trunk strength and control (for example, side bridge test).

Key outcome measures

A patient-reported outcome measure (PROM) is the gold standard when evaluating an outcome in a clinical population (Chapter 16). When choosing a PROM, the psychometric properties of all possible PROMs should be considered, specific to the patient. Other questionnaires such as the Hip Outcome Score (HOS)⁵⁵ and the Modified Harris Hip Score⁵⁵ are not recommended as their validity and responsiveness have not been established.

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We recommend the Copenhagen Hip and Groin Outcome Score (HAGOS),⁵² the Hip Dysfunction and Osteoarthritis Outcome Score (HOOS)⁵³ or the international Hip Outcome Tool-33 (iHOT-33)⁵⁴ (Table 31.3).

Investigations

Plain radiography, MRI and ultrasonography are the mainstay of imaging for diagnosis of hip pathology. Plain radiographs are often overlooked by clinicians, but usually should be the first investigation ordered as they can provide valuable information. FAI can often be seen on an anteroposterior view or a lateral view of the pelvis, such as a Dunn view (Fig. 31.7), cross-table view or frog-leg view

Table 31.3 PROMs for hip pain

Copenhagen Hip and Groin Outcome Score
• Self-administered questionnaire evaluating symptoms and function in patients with hip and groin pain • Six subscales scored separately: symptoms, pain, activities of daily living, sport and recreation, physical activities and quality of life • Raw scores converted to score from 0 to 100 (100 = best possible score) • Test–retest reliability: ICC 0.82–0.91 • MDC95%: 8–19 points • MCID: 6–10 points • Available online: www.koos.nu
Hip Dysfunction and Osteoarthritis Outcome Score
• Self-administered questionnaire evaluating symptoms and function in patients with hip arthroscopy, hip osteoarthritis and intra-articular causes of hip pain • Five subscales scored separately: pain, other symptoms, daily living, sport and recreation, and quality of life • Raw scores converted to score from 0 to 100 (100 = best possible score) • Test–retest reliability: ICC 0.93–0.96 • MDC95%: 9–17 points • MCID: 6–11 points • Available online: www.koos.nu

continued

Table 31.3 Cont.

The international Hip Outcome Tool-33
• Self-administered questionnaire evaluating symptoms and function in younger, active patients with a variety of hip pathologies • Thirty-three items • Each of the 33 items scored out of 100. Points summed and divided by number of items to get an overall score out of 100 (100 = best possible score) • Test–retest reliability: ICC 0.78 • MDC95%: 16 points • MCID: 6 points • Available as appendix to Mohtadi et al. 2012⁵⁴



Figure 31.7 Dunn 45° flexion view indicating sclerotic cam lesion at femoral head–neck junction (arrow), pincer impingement at acetabulum and bony cyst at femoral head–neck junction

of the hip. Similarly, these plain radiographs can also guide the clinician to the presence of OA and abnormalities in morphology such as acetabular dysplasia, acetabular retroversion or anteversion, the presence of os acetabulare, and not-to-be-missed pathologies such as a slipped upper femoral epiphysis, Perthes' disease, tumours, fractures of the neck of femur and avascular necrosis. Unfortunately, a plain radiograph does not provide information about soft tissue injuries such as labral, chondral, ligamentum teres or tendon pathology.

MRI is commonly used in the diagnosis of soft tissue injuries of the hip. Pathologies such as labral tears, ligamentum teres tears, tendon and bursae pathology, and occasionally chondral defects may be seen on an MRI scan. In recent years, the sensitivity and specificity of MRI to these injuries has improved (see Table 31.4), although false-negative results are often noted. Therefore, imaging findings should be taken in combination with the patient's history and physical examination in order to obtain a correct diagnosis.

Due to the depth of the joint, diagnostic ultrasound is not especially useful for the diagnosis of intra-articular hip

pathology; however it can be useful in determining the presence of bursae of the greater trochanter or iliopsoas tendon, and tendinopathy of these regions. Real-time ultrasound can be used to assess the function of the deep lumbar and hip stabilising muscles.

Diagnostic injections of local anaesthetic are used frequently in the hip to determine the presence of intra-articular pathology of the hip. These are generally performed under imaging guidance. A reduction in symptoms following an injection generally confirms the presence of intra-articular pathology, although a negative response does not necessarily indicate that no pathology is present, and further investigation and management may still be warranted.

Surgeons may obtain specific computed tomography (CT) scans pre-operatively to assist them in planning surgery for the treatment of FAI, but these are not routinely used in clinical practice for diagnosis of the source of hip pain.

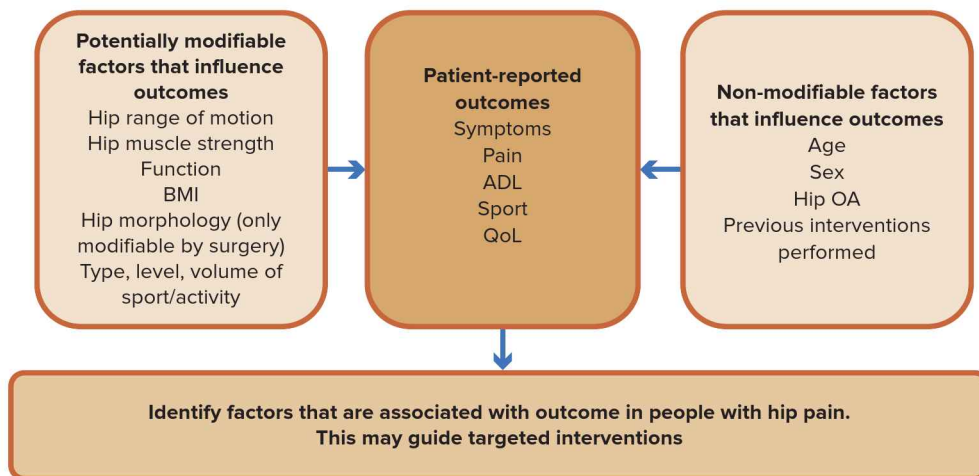
PREDISPOSING FACTORS FOR HIP PAIN

A number of intrinsic and extrinsic factors can contribute to hip region pain (Fig. 31.8). Such factors may be modifiable or non-modifiable. Modifiable factors should be targeted for treatment as they can alter the loads on the hip joint.

Extrinsic factors may include the type of sport played, particularly those involving repeated combined hip flexion, abduction and adduction, and loaded rotational or twisting movements. These factors may also include the volume of sport and activity undertaken, footwear worn or type of surface played upon. Of these extrinsic factors, the type of sport and volume of load undertaken are probably the most important when evaluating the athlete with hip pain. Sport and activity involving repeated hip flexion, abduction and adduction; and rotation and twisting are reported throughout the literature as influencing the likelihood of the development of hip pathology. The clinician must examine these loads in

Table 31.4 Diagnostic accuracy of imaging tests in patients with hip pain

	Sn	Sp	LR–	LR+
Gluteal tendinopathy				
Ultrasound	79–100	100		
MRI ⁵⁶	33–100	92–100		
Labral tear				
Ultrasound ^{57, 58}	82–94	0–60	0.30	2.05
CT ^{59, 60}	92–97	87–100		
Multidetector CT (MDCT) ^{61–64}	88–100	87–100		
MRI ⁶⁵	66	79		
MRI + contrast (magnetic resonance arthrography—MRA) ^{65, 66}	87	64		
Cartilage				
CT ^{59, 67, 68}	66–88	40–94		
MRI	59	94	0.44	9.83
MRA ^{66 59, 68–71}	49–97	33–100		
FAI				
MRI ⁶⁵	66	79	0.43	3.14
MRA ⁷²	91	80	0.11	4.55
CT	92–97	87–100		

**Figure 31.8** How modifiable and non-modifiable factors contribute to hip-related symptoms

detail and modify accordingly for those athletes that are experiencing hip pain.

Intrinsic factors can also influence the development of hip pain and pathology. These factors may also alter loads within the joint, predisposing the hip to injury. Intrinsic

factors are either 'local', 'remote' or 'systemic', and must be considered for comprehensive assessment of the athlete with hip pain. Identifying these factors via a thorough assessment is essential if the clinician is to successfully modify the loads within the joint to protect potentially

vulnerable structures. An understanding of all possible contributing factors to hip pain will assist the clinician in devising a targeted treatment plan.

Local factors

A number of modifiable local factors can contribute to hip pain and are potential targets for treatment, as shown in Table 31.5.

Remote factors

Proximal and distal remote factors may contribute to the development of hip pain. Those factors that are modifiable should be targeted when planning treatment programs for patients with hip-related pain.

Proximal factors

Increased pelvic tilt and/or lumbar hyperextension may increase the load on the anterior margins of the hip, due to the more distal placement of the anterior acetabular rim. Patients with increased pelvic tilt and

reduced trunk strength can have increased acetabular retroversion and reduced femoral internal rotation range.⁷³ This increased load may be a source of increased hip pain and eventually anterior hip pathology. The clinical assessment of pelvic symmetry and lumbar spine is outlined in Chapter 29.

Inadequate control of the lumbopelvic segments may result in a number of asymmetries, which alter the loads on the hip joint. In particular, lateral pelvic tilt may increase the load on both the lateral and medial structures of the hip joint, due to the increased adductor and internal rotation moment seen on the stance leg.

Hip and lumbopelvic control can be assessed using the single-leg squat (Chapter 8), other single-leg activities, gait or sport-specific activities. In some cases, the athlete should also be videoed whilst running, particularly when fatigued, as altered control may become more pronounced. The demands on the lumbopelvic region for the individual's sport must be considered, as this may predispose certain athletic groups to fatigue and subsequently altered load on the hip joint.

Table 31.5 Local factors that can contribute to hip-related pain and their possible mechanisms

Factor	Possible mechanisms	Confirmatory assessments
Reduced hip flexion range of motion	Structural • Cam and/or pincer impingement • Chondrolabral pathology	Plain radiograph • Anteroposterior pelvis • Dunn view 45°/90° Positive FADIR (Fig. 31.6b) MRI
Reduced hip internal rotation range of motion	Structural • Acetabular retroversion • Cam and/or pincer impingement • Femoral retroversion • Osteoarthritis changes (osteophytes)	Plain radiograph • Anteroposterior pelvis • Dunn view 45°/90°
	Functional • Reduced strength hip internal rotators • Tight gluteals and piriformis • Muscle spasm	Hand-held dynamometry Muscle-length tests
Increased hip rotation range (external rotation and/or internal rotation)	Structural • Acetabular anteversion • Acetabular dysplasia • Capsular laxity	Plain radiograph
Altered movement patterns	Reduced muscle strength • Reduced hip abductor, adductor, extensor strength, external rotator strength Muscle inhibition, disuse atrophy Reduced performance in single-leg squat, neuro-motor control and balance tasks	Hand-held dynamometry Palpation, observation, real-time ultrasound Functional and balance tests

Distal factors

Increased subtalar pronation may lead to an increase in tibial internal rotation. This in turn may lead to an overload on the iliotibial band (ITB) and the lateral structures of the hip. Increased ITB tension results in increased compression over the greater trochanter and the development of gluteus medius and minimus tendinopathy and trochanteric bursitis. Increased tibial internal rotation may also create increased internal rotation of the femur, thus heightening load on the hip, particularly in athletes with an increased risk of impingement. The clinical assessment of the subtalar joint is outlined in Chapter 41.

Adequate range of ankle dorsiflexion during the stance phase of gait is essential in order to minimise excessive loads further up the kinetic chain. In addition, reduced ankle dorsiflexion range can increase the amount of adduction at the knee in single-leg squatting tasks, therefore potentially increasing hip impingement. If this movement is limited, the gait pattern may be altered to achieve onward forward propulsion of the athlete. One adaptation commonly seen is an increase in hip adduction and rotation at the middle of the stance phase of gait. This may increase load on the hip joint in similar ways outlined above. The clinical assessment of ankle dorsiflexion range of motion is outlined in Chapter 41.

Systemic factors

Common 'systemic' factors contributing to hip joint loads include patient age, sex, BMI and family/genetic predisposition. Of these, BMI is the only factor that is potentially modifiable. Patients with a BMI >25 should be advised to seek appropriate guidance to reduce BMI to a healthy level. Other non-modifiable factors should be considered when considering a patient's likely outcomes, prognosis and expectations, as well as appropriate surgical versus non-surgical treatment options.

FEMOROACETABULAR IMPINGEMENT

Femoroacetabular impingement (FAI) is defined as a pathological mechanical process by which morphological abnormalities of the acetabulum and/or femur combined with vigorous hip motion can damage the soft tissue structures within the hip joint itself.⁷⁴ For the clinical diagnosis of FAI, the hip should be symptomatic. There are two types of FAI which differ in both the anatomical site of the morphological abnormality and the mechanism by which it causes intra-articular damage.⁷⁵

Types of FAI—cam and pincer impingement

The first FAI type is 'cam impingement', in which the morphological abnormality is located on the femoral side

of the hip joint. The cam deformity is extra bone formation at the anterolateral head-neck junction, causing a non-spherical femoral head.¹¹ During hip motion, particularly flexion and internal rotation, the cam deformity can be forced into the acetabulum, causing shear forces at the chondrolabral junction (Fig. 31.9b).⁸

The second type of FAI is referred to as 'pincer impingement' (Fig. 31.9c). This refers to a morphological or orientational abnormality of the acetabulum leading to an over-coverage of the femoral head. This can either manifest as a deep acetabulum or as a retroverted acetabulum, which are most commonly seen anteriorly. Particularly during hip flexion, the femoral neck can impinge against the over-covered acetabulum, causing an impaction on the labrum. This type of impingement might lead to labral and also cartilage damage throughout the acetabulum in a small thin strip around the labrum.⁸

A cam deformity and a pincer deformity may exist together and when this leads to symptoms it is referred to as a mixed type of FAI. It is very important to recognise that a cam deformity or a pincer deformity will not always cause FAI. Susceptibility is highly dependent on many co-factors including other anatomical characteristics of the hip such as femoral and acetabular version, on the type and intensity of hip movement, and on the vulnerability of the labrum and cartilage itself.⁷⁶

Prevalence of FAI

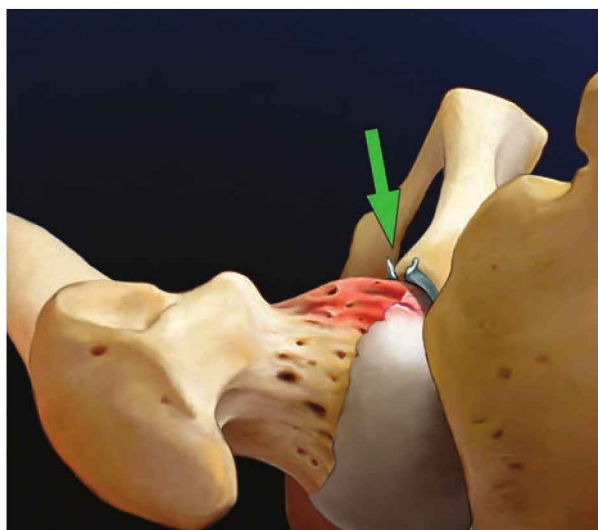
As the mechanical impingement seen in FAI is difficult to quantify, most studies on the prevalence of FAI have



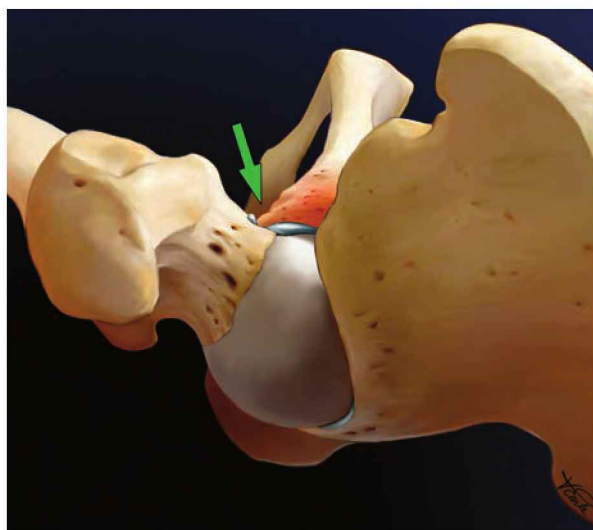
(a)

Figure 31.9 Hip joints with and without FAI

(a) Hip joint without FAI



(b)



(c)

Figure 31.9 (cont.) (b) Cam lesion—additional bone arises as a ‘bump’ on the femoral surface (c) Pincer lesion—bone spur extends from the acetabular surface

focused on the morphology, that is, the presence of a cam or pincer deformity. It is important that the clinician understands that the morphology that can lead to FAI is highly prevalent and is in itself not a pathology.

The prevalence of cam deformity in the general asymptomatic population has been estimated at around 10–25%.^{6, 77} Cam deformity is more prevalent in males (25–50%) than in females (0–10%).⁶ Interestingly, the prevalence can be extremely high in male athletes—up to 89%—compared to only 9% in non-athletic controls.⁷⁸

Pincer deformities are generally more prevalent in women, but the prevalence varies widely between studies because of the heterogeneity in what is considered a pincer deformity. General over-coverage is a condition in which the femoral head is positioned deep in the acetabulum or when there is an overgrowth of the acetabular rim. This type of pincer deformity is present in about 20% of the population.^{79, 80} When focal over-coverage or acetabular retroversion is also taken into account, the prevalence can be as high as 60%.⁸¹ Whether the latter should be regarded as a pincer deformity is still under debate.⁸² It is unknown if pincer deformity is more prevalent among athletes.

Aetiology

Hip loading during adolescence is emerging as a key factor in the aetiology of a cam deformity. Cam deformity was first recognised by Murray et al. in 1971,⁸³ and more recently,⁷⁸ hip loading during skeletal maturation has appeared to be related to development of a cam

deformity (Fig. 31.10a and b). Since 2011, two cohort studies described the development of a cam deformity during adolescence.^{78, 84} The prevalence was higher in both football players and basketball players than in non-athletic controls.^{78, 84} In these young athletes, the extra bone formation in the anterolateral head-neck junction becomes radiographically visible from around the age of 13 years and the cam deformity develops gradually until the growth plate closes.⁸⁵ After growth plate closure, the morphology of the proximal femur appears to change minimally.⁸⁵ This is supported by the high prevalence of cam deformity in adult athletes participating in high-impact sports ranging from 60% to 89%.^{78, 81, 86} Therefore, a cam deformity is most likely a bone adaptation in response to vigorous hip loading when the growth plate is still open. During the time of skeletal maturation, there might be a dose-response relationship between the intensity of sports practice and the development of a cam deformity. Genetics might also play a role, as a cam deformity might have a familial pattern, with siblings being three times more likely to have a cam deformity than controls.⁵⁰ The aetiology of a pincer deformity is unknown and there are no studies available that investigate how a pincer deformity develops.

Association with pain and pathology

A prerequisite of the FAI definition is the presence of hip pain and symptoms. Without hip pain, the diagnosis of FAI cannot be made. The presence of a cam or a pincer deformity in isolation have been associated with hip pain



(a)



(b)

Figure 31.10 (a) Development of a cam deformity over time during skeletal maturation (b) Extension of the growth plate towards the femoral neck during the development of a cam deformity

in young adults,⁸⁷ but only 26% of people with a cam deformity complain of hip pain.⁸⁸ Asymptomatic young adults with a cam deformity have a 4.5 times higher risk of developing hip pain within 4 years than those without a cam deformity. This was shown in a cohort of 170 subjects, in which 16% of asymptomatic hips with a cam deformity at baseline had at least one episode of hip pain lasting longer than 6 weeks within the next 4 years.³

However, athletes might have a higher risk of developing hip pain resulting from a cam deformity, particularly when they experience repetitive movements of flexion and internal rotation of the hip. In American football (gridiron) players, the presence of a cam deformity is associated with greater hip and groin symptom prevalence, whereas a pincer deformity is not.⁸⁹ In ice hockey players, hip pain together with symptoms of FAI is associated with a cam deformity at a mean age of 19 years.⁹⁰ The association between a cam deformity and hip pain is likely also true for other sports requiring movements of deep flexion, rotation and pivoting.

Following the anterolateral location of a cam deformity, the anterosuperior region of the acetabulum is mostly affected corresponding to the site where the cam deformity is forced into the acetabulum with

PRACTICE PEARL

Cam impingement is associated with intra-articular pathology, whereas the association between pincer impingement and intra-articular pathology is less clear.

flexion and internal rotation.⁸ This results locally in high shear forces at the acetabular chondrolabral junction. Resultant damage to the labrum can progress to its detachment from the acetabular rim and involvement of the acetabular cartilage. This can range from slight softening and swelling of the cartilage in the early stage, to delamination of the cartilage from the subchondral bone in later stages, a so-called carpet lesion. The size of the cam deformity is positively correlated with more severe cartilage damage.⁹¹

Severe intra-articular hip damage is already seen in athletes around their twenties. Even in asymptomatic males with a mean age of 20 years, those with a cam deformity are 2.8 times more likely to have labral lesions and a decreased combined femoral and acetabular cartilage thickness of 0.19 mm.⁶ In the presence of a pincer deformity, labral lesions and a more circumferential pattern

of acetabular cartilage damage have been described based on intra-operative findings, but epidemiological data on intra-articular hip pathology in those with asymptomatic pincer deformities is lacking.⁸

Osteoarthritis

A cam deformity is associated with hip OA, whereas the association between a pincer deformity and OA is conflicting. Since a cam deformity is present immediately after skeletal maturity, prospective studies to evaluate the risk for developing OA would require a long-term follow-up of decades. Therefore, current cohort studies consist mostly of individuals aged over 40 years. In this population, there is strong evidence that a cam deformity is a risk factor for development of OA.^{76, 92} A prospective study found that a cam deformity conferred a four times higher risk of developing OA within 5 years and up to 10 times higher risk with a greater cam deformity.⁹³ Individuals with a cam deformity and decreased internal rotation $\leq 20^\circ$ had a 53% chance of developing OA compared to only 2% in people without these features. Most other case-control, cross-sectional and retrospective cohort studies also show an increased risk for OA.^{79, 94, 95} Regarding the higher prevalence of cam deformity in athletes and the higher chance to experience FAI in the presence of a cam deformity because of the repetitive movements they undertake, athletes might even be more likely to develop OA due to cam-type FAI.

The relationship between pincer deformity and OA is less clear. Some studies show a moderately increased risk for having OA in the presence of a pincer deformity,⁷⁹ while others, including higher-quality studies, showed no association or even a protective effect of a pincer deformity on development of OA.^{94, 96}

PRACTICE PEARL

This association between FAI and increased incidence of other hip pathology has resulted in considerable debate regarding the benefit of prophylactic surgery to correct deformity associated with FAI in athletes who do not have hip or groin pain. As there is no conclusive evidence at this stage which indicates that all athletes with FAI will develop other hip pathology, prophylactic surgery to correct deformities in athletes who do not have signs of hip pathology is not recommended.

The early identification of FAI in athletes with hip and groin pain is essential. Unfortunately there is no gold standard for clinical diagnosis of FAI. Clinical signs that are often reported to indicate the presence of FAI include

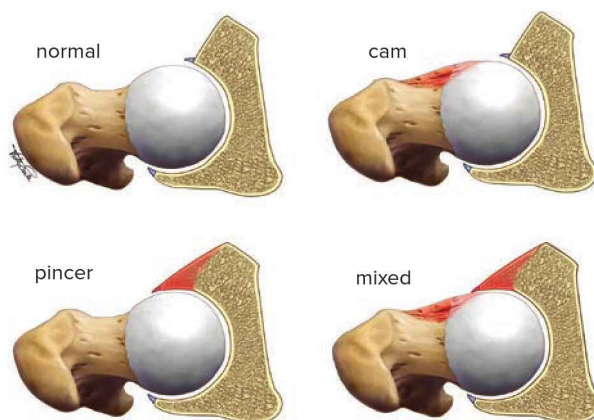


Figure 31.11 Appearances of the different types of FAI

reduced range of hip internal rotation, particularly when the hip is flexed, and a positive FADIR test. Unfortunately these tests are not specific to detect FAI and may result in a high number of false-positive results. Therefore, radiological examination is required.⁴² Plain radiographs can be useful and generally a plain anteroposterior view of the pelvis will indicate the presence of FAI when read by an experienced radiologist (Fig. 31.11).

Athletes who present with FAI should be encouraged to avoid the position of impingement as much as possible. This position of impingement is usually flexion, internal rotation and adduction or any combination of these (Fig. 31.12). This may involve activity modification on a day-to-day basis, as well as during athletic pursuits. For example, in ball sports this may involve playing in a different position which requires less time changing direction and getting down low to the ball. It may also involve reducing the time spent on the field. Maximising dynamic neuromotor control around the hip will also assist in achieving this goal.

LABRAL TEARS

Acetabular labral tears (Fig. 31.13) are seen frequently in the athletic population, with 22% of athletes with groin pain having labral tears, and 55% of patients with mechanical symptoms and hip pain having labral tears.^{18, 22, 44, 97} The aetiology of labral tears is well described in the literature. However, studies have also shown that there is a high prevalence of labral tears in the asymptomatic population, especially as people age, and as with many other parts of the musculoskeletal system, pain does not always match pathology.

Pathology

The presence of both cam-type FAI^{5, 8, 11, 98, 99} and developmental dysplasia of the hip (DDH)^{100, 101} can increase the risk of a labral tear. Mechanisms include impingement

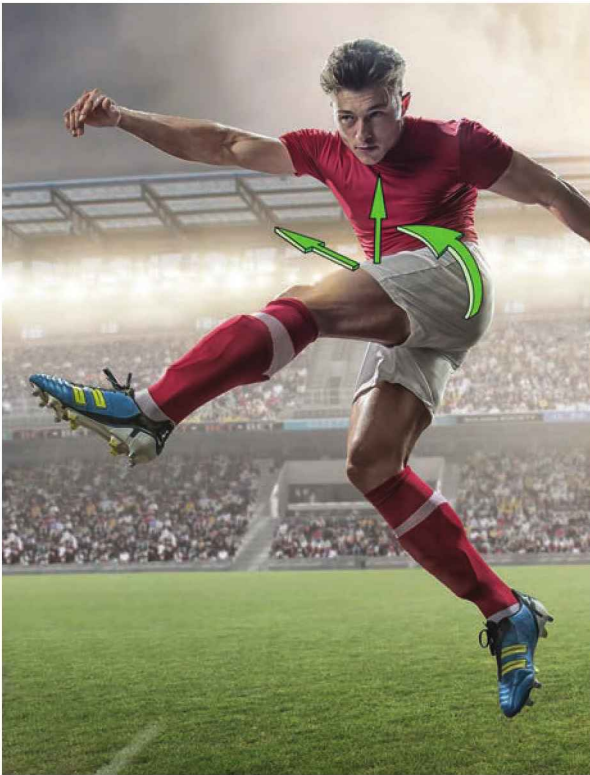


Figure 31.12 Hip impingement during kicking motion—flexion, internal rotation and adduction
ISTOCK/VICKY EARLE

of the labrum in the presence of FAI and increased shear forces on the outer joint margins, including the labrum, in the presence of DDH. In addition, patients with labral tears are 40% more likely to have coexisting chondropathy.¹⁴ The inability of a damaged labrum to adequately sustain chondral nutrition via the synovial fluid, as well as the reduced load-bearing capacity of a damaged labrum, may provide an explanation for these findings.

The prevalence of labral tears in the USA and Europe is greatest anteriorly.^{18, 22, 97, 102} Various causes for the high number of anterior labral tears have been postulated, including reduced thickness of the labrum anteriorly, the prevalence of FAI lesions seen anteriorly resulting in anterior impingement, and common functional activities, especially those with repetitive twisting and pivoting of the hip.⁵ The reduced bony support seen anteriorly in the hip due to the anteverted position of the acetabulum, which results in higher shear forces on anterior soft tissue structures, is also a likely cause of labral pathology. It has been shown that in the last 20–30% of the stance phase of gait and in more than 5° of hip extension, increased forces are placed on anterior soft tissue structures by the head of femur.^{18, 103}

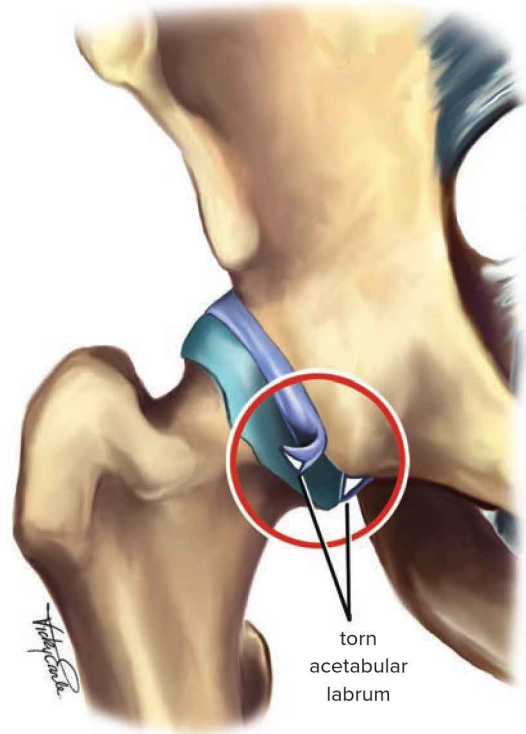


Figure 31.13 Labral tear

Tears of the acetabular labrum are usually classified as type I or type II tears.^{22, 104} Type I is described as a detachment of the labrum from the articular hyaline cartilage at the acetabular rim. Type II is described as cleavage tears within the substance of the labrum. The location of these tears relative to the vascularisation of the labrum influences the potential for healing of the tear and the most appropriate type of intervention. Recently it has been suggested that the innervation of the labrum may allow for the capacity for regeneration and repair.²⁰

Clinically, the identification of labral tears in patients remains difficult. The patient often complains of mechanical symptoms such as locking, clicking, catching and giving way. The location of pain is usually reported to be within the anterior hip or in the anterior groin region, although some patients report pain in the posterior buttock.

Clinical examination is also difficult. While clinical tests such as FADIR (Fig. 31.6b) and FABER (Fig. 31.6c) have high sensitivity, meaning a negative test will rule out intra-articular hip pathology, they have low specificity, meaning they are not good at determining which intra-articular hip structure is the cause of pain (Table 31.2). The Thomas test (Fig. 31.6d) has a high likelihood ratio, where a positive test (pain on testing) is likely to indicate the presence of intra-articular hip pathology (Table 31.2). This may also reflect the common situation of multiple intra-articular



Figure 31.14 T1 fat suppressed coronal MR image showing an anterior-superior labral tear (arrow)

hip pathologies coexisting. The diagnostic accuracy of radiological investigations for labral tears has improved in recent years, with MRI and MRA (magnetic resonance arthrography) (Fig. 31.14) both having a reasonable degree of sensitivity and specificity (Table 31.4).^{40, 66} However, labral tears may be suspected but not confirmed until patients present for hip arthroscopy, which remains the gold standard to diagnose labral pathology.^{18, 22}

Athletes with labral pathology may respond to conservative management and this should always be trialled prior to undergoing surgery. Management should be directed to unloading the damaged labrum which is almost always anterior and/or superior. Repetitive hip flexion, adduction or abduction and rotation at the end of range should be avoided through activity modification. Improving hip joint neuromotor control via activation of the deep stabilising muscles, initially in an unloaded then a progressively loaded manner, appears to assist in the unloading of the labrum.

Gait retraining may also be undertaken to minimise excessive hip extension at the end of stance phase of gait, as increased hip extension has been demonstrated to increase the loads on anterior hip joint structures.¹⁰⁵ Neuromotor control of the hip should be maximised as outlined above, and any remote factors influencing the mechanics of the hip should be addressed.

LIGAMENTUM TERES TEARS

Ligamentum teres tears (Fig. 31.15) are commonly reported in athletes undergoing hip arthroscopy and are being reported more frequently in the literature. Studies have found up to 70% of athletes undergoing hip arthroscopy for FAI and labral tears also have tears of the ligamentum

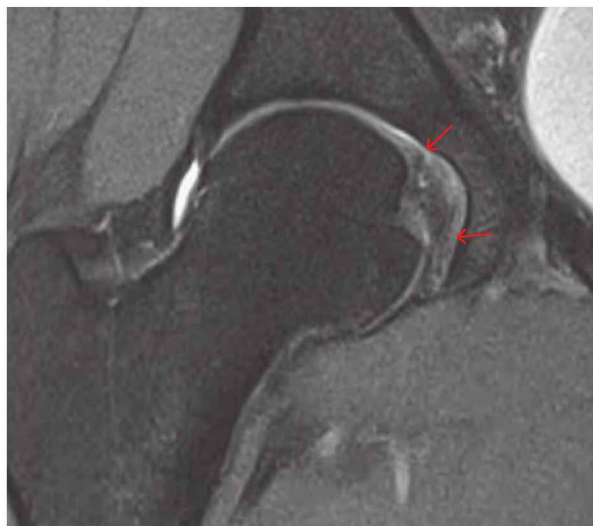


Figure 31.15 Coronal T2-weighted MR arthrogram of a hypertrophied, irregularly thickened, partially torn ligamentum teres that carries greater signal than a normal dark ligament (between arrows)

teres.⁵ Tears of the ligamentum teres are classified as type I which is a partial tear, type II which is a complete rupture and type III which describes a degenerate ligament.¹⁰⁶ Ligamentum teres tears can occur in isolation, but often coexist with FAI, dysplasia and synovitis, probably due to the altered joint loads seen in these conditions.

The mechanism of injury for ligamentum teres most commonly involves forced flexion and adduction, and often internal or external rotation.¹⁰⁶ Twisting motions and hyperabduction injuries have also been reported to cause a tear to this ligament. With the likelihood of the ligamentum teres playing a large proprioceptive and stabilisation role of the hip becoming increasingly recognised, the prompt diagnosis and management of these injuries in the athlete is essential. Likewise, any surgical procedure that sacrifices the ligamentum teres through open dislocation should be carefully considered.

If suspected, the principles of management are similar to those of labral pathology, with a particular emphasis on regaining neuromotor control, excellent proprioception and avoiding positions that place the ligament under most stress using activity modification.

SYNOVITIS

Synovitis (Fig. 31.16) is often seen in athletes with other intra-articular hip pathology, whether it is FAI, labral tears, ligamentum teres tears or chondropathy. It is rarely seen as a primary entity. Synovitis can cause considerable pain in the hip joint, with night pain and pain at rest being common presentations.

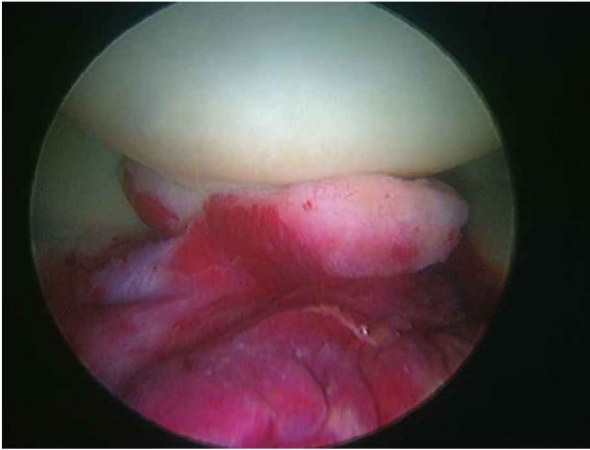


Figure 31.16 Synovitis

Synovitis is a concern to the clinician due to the pain and associated changes in muscle activation that are seen around the hip in the presence of pain. In addition, the implications of synovial dysfunction on cytokine production, nutrition and hydration of articular cartilage, which may already show signs of chondropathy, is significant for the long-term health of the hip joint.

Management should aim to address the other coexisting pathology, restore normal neuromotor control around the hip, modify loads as well as introduce anti-inflammatory measures such as oral non-steroidal anti-inflammatory drugs (NSAIDs) or intra-articular injections. Peak synovial fluid pressure is reduced in positions of hip flexion and internal rotation, already reducing potential nutrition to chondral surfaces.²⁶ When synovitis is present, these positions should be avoided to try and optimise movement of synovial fluid through the joint.

CHONDROPATHY

Changes to the chondral surfaces of the hip are often seen in conjunction with other hip pathologies (Fig. 31.17). It is well reported that the presence of FAI,^{8, 9, 13, 14, 107} decreased acetabular anteversion,¹⁰ labral pathology^{14, 18, 101} and DDH^{9, 10, 101} can increase the risk of chondropathy and ultimately OA of the hip. In patients with significant labral pathology, chondral loss is often up to 70% of the full thickness or Outerbridge grade III or IV.¹⁰⁸ It is also proposed that the presence of longstanding synovitis may also affect the nutrition of chondral surfaces, possibly exacerbating chondral damage.

The presence of chondropathy at arthroscopy is associated with worse patient-reported outcomes, up to 3 years post-operatively, especially compared to healthy controls. In addition, outcomes are unlikely to improve over time.¹⁴ Patients with severe chondral



(a)



(b)

Figure 31.17 (a) Coronal MR arthrogram showing full thickness chondral defect in the central acetabulum (b) Arthroscopic view of severe chondral damage secondary to FAI

lesions are more likely to have worse outcomes than those with mild lesions¹⁴ and have a much higher risk of progressing to total hip replacement within 2 years of an arthroscopic procedure.¹⁰⁹ Outcomes for people with severe chondral lesions look similar to those reported for people with advanced hip OA,¹⁴ and it is now thought that chondropathy may represent hip OA in its early stages.

The majority of chondral lesions are seen on the anterior or superior aspect of the acetabular rim, at the chondrolabral junction. This is not surprising considering

that this is also the location for the majority of cam and pincer lesions and the majority of labral tears (see above). DDH also involves a reduction in the bony coverage of the femoral head by the acetabulum, thus the anteriorly directed forces of the femoral head will be concentrated on a smaller surface area on the anterior aspect of the joint.

The clinical diagnosis of chondropathy is unlikely to be confirmed with plain radiographs because early chondral changes will not be visible, unless the disease is well advanced. MRI may identify earlier chondral lesions, although the extent of chondropathy is often only evident on hip arthroscopy.¹⁴ The presence of FAI, labral tears and DDH should also increase the suspicion of chondropathy as these often coexist. Clinical tests such as FADIR, which have high sensitivity for intra-articular pathology, may increase suspicion of a chondral lesion; however, this is often only confirmed arthroscopically.

Chondropathy is difficult to manage and may be difficult to confirm in the early stages without arthroscopic confirmation. If suspected, the management again is similar to that of labral pathology, as the majority of chondral lesions of the hip occur in the anterior aspect of the acetabular rim at the chondrolabral junction. As such, this region should be unloaded in the same fashion as labral pathology, with an emphasis on regaining normal neuromotor control of the hip. Recent evidence has shown atrophy in the inferior gluteus maximus and hypertrophy in gluteus medius in early OA, with atrophy also occurring in gluteus medius in severe OA.^{110, 111} As chondropathy is likely to represent hip OA in its early stages, aspects of treatment that have been described as being effective for hip OA should be introduced. This may include weight reduction where appropriate, lifestyle factors, general fitness programs, aquatic exercise programs and specific strengthening exercises for hip muscles.

Attempts to minimise synovitis should also be made as the synovium and synovial fluid plays an important role in articular cartilage nutrition. As outcomes of hip arthroscopy for individuals with significant chondral loss are generally worse than for those with no chondral loss and the risk of hip replacement is high,^{5, 109, 112-114} conservative measures should be attempted first and in some cases the athlete counselled to modify the amount of weight-bearing activities they undertake.

HIP INSTABILITY

Hip instability is increasingly being recognised as an important potential source of hip pain. Gross instability of the hip is seen in DDH and generalised hypermobility, while localised directional instability may be present in cam-type FAI (as a contra-coup lesion), ligamentum teres tears

(due to reduced proprioception), hips with large amounts of anteversion (as capsular insufficiency anteriorly) and in 'normal' hips that are subjected to extremes of range of motion, such as dancers and gymnasts. There is no good clinical or radiographic test for instability and diagnosis is usually one of clinical suspicion when extremes of range of motion, poor balance, poor control in functional tasks, observable excessive translation of the femoral head, and clicking and clunking are present. Treatment for hip instability involves restoring hip muscle strength, improving balance and functional control, modification of functional tasks to avoid extremes of range, and education regarding the risks of repeated activity in unsafe and extremes of range. Some surgeons have reported treating hip instability by arthroscopic shortening of the hip capsule, but outcomes for this procedure are unknown. In extreme cases of DDH and abnormal acetabular version, patients may consider major orthopaedic procedures such as peri-acetabular osteotomy.

TREATMENT OF HIP IMPAIRMENTS

This section will describe the impairments associated with hip pathology and explain how to treat them. The subsequent section will then outline the application of those same principles of management to patients post hip arthroscopy.

Principles of rehabilitation of the injured hip

Rehabilitation of the injured hip requires careful consideration of the interplay between pain and loading (including progression of exercises and activities). Importantly, due to its role in all activities of daily living, including simple activities such as sit-stand, standing and walking, it is hard to 'rest' the hip. Given this, it appears the most provocative loading for the hip occurs when rotational loads, loading at extremes of range and overload in the impingement position occur, rather than excessive amounts of walking, running or other weight-bearing activities that do not involve extremes of rotation, impingement and other end-range positions. It is vital that the patient and the clinician have a good understanding around monitoring joint loads and the loading response.

Unfortunately, there is no level I or II evidence that supports the ideal rehabilitation program or evaluates the effectiveness of particular principles of rehabilitation of the hip. However, the general principles of management of hip pathology are straightforward and are consistently reported in clinical commentaries available concerning rehabilitation of the hip.^{5, 18, 115-118} In addition, several recent studies have detailed the physical impairments that exist in people with hip pain and pathology, around which an impairment-based rehabilitation guideline can be based.

The most commonly reported physical impairments seen in hip pain and pathology:

1. reduced hip joint range, especially flexion and internal rotation^{15, 119, 120}
2. reduced hip muscle strength in all hip muscles, especially abductors and adductors (men and women), extensors (mostly women) and external rotators (mostly women)^{15, 119, 121, 122}
3. reduced balance in single-leg dynamic tasks¹²³
4. increased femoral adduction in single-leg squats
5. reduced trunk muscle strength
6. alterations in gait biomechanics (primarily reduced range of hip motion in gait)^{124, 125}
7. reduced functional performance, especially in hopping and squatting tasks¹⁵
8. probably adverse loading within the hip due to morphology, changes in range of motion, hip muscle weakness and poor functional performance
9. abnormalities of the kinetic chain.

Nine principles of rehabilitation for hip pain patients

1. Restore hip range of motion

Hip range of motion may impact on load within the hip. Hip flexion^{35, 127} and hip extension¹²⁷ may increase load on the anterior^{103, 127} and superior³⁵ regions of the hip. As most FAI lesions and associated hip pathology occur in the anterior and superior regions of the hip,^{41, 88} loads associated with these regions of the hip require consideration. Range of motion may be limited in this group if movement at the end of range loads damaged tissue in a manner that provokes pain. Often a patient's range of motion is limited to protect damaged regions of the hip and, as such, gaining greater range of motion may also worsen symptoms. Manual techniques such as soft tissue release and needling, stretching and muscle activation may improve range, but should be done with respect to the athlete's pain during and after treatment.

B

Table 31.6 Best available evidence reports of impairments seen in patients with hip pain

Impairment	Deficit observed	Level of evidence**	References
Range of motion			
	↓ Flexion range of motion [†]	3b	15
	↓ Abduction range of motion [†]	3b	15
	↓ Internal rotation range of motion in 90° [†]	4	119
	↓ External rotation range of motion in 90° ^{††}	3b, 4	15, 119 (Two only in females)
	↑ Extension range of motion [†]	4	119
Strength			
	↓ Flexion strength ^{†Ω}	3b, 4	15, 126
	↓ Extension strength [†]	4	119
	↓ Abduction strength ^{†‡✕}	3b, 4	15, 119, 122
	↓ Adduction strength ^{†‡}	3b, 4	15, 119
	↓ Internal rotation in 0° strength [✕]	4	122
	↓ External rotation in 0° strength [✕]	4	122
	↓ Internal rotation in 90° strength ^{†✕}	4	119, 122
	↓ External rotation in 90° strength ^{†‡✕}	3b, 4	15, 119, 122
Neuromuscular activity			
	↓ Tensor fasciae latae activity in hip flexion maximal voluntary contraction	3b	15
Gait			
	↓ Frontal plane hip range of motion (abduction/adduction) [†]	3b	15
	↓ Sagittal plane hip range of motion (flexion/extension) [†]	3b	15
	↓ Internal rotation range of motion [†]	3b	15

continued

Table 31.6 Cont.

Impairment	Deficit observed	Level of evidence**	References
Activity limitations			
	↓ Squatting depth	3b	15
	↓ Sagittal plane range of motion when ascending stairs	3b	15
	↓ Maximum hip extension range of motion when ascending stairs	3b	15
	↓ Internal rotation range of motion when ascending stairs	3b	15
Balance			
	↑ Medial lateral range in single-leg squats†	4	123
Patient reported outcomes			
	↑ Pain§	4	38
	↑ Stiffness§	4	38
	↓ Sports participation§	4	38
	↓ Quality of life§	4	38
	↓ Activities of daily living§	4	38

† chondrolabral pathology; ‡ chronic hip joint pain; † symptomatic femoroacetabular impingement; § severe chondropathy; Ω labral pathology; ** NHMRC levels of evidence

Blue = supported by meta-analyses (systematic reviews) or randomised controlled trials

Orange = supported by nonrandomised studies and case series

Black = supported by expert opinion, clinical guideline

Red = current evidence does not support

2. Restore hip muscle strength

Restoration of hip muscle strength should follow these principles.

Phase 1: Deep hip stabiliser retraining

The SHS muscles are those with the greatest capacity to provide dynamic stabilisation of the hip (see above). Retraining of these deep hip stabilisers may be undertaken in the early stages of rehabilitation. As with other pain conditions,¹²⁸ clinical observation indicates that pain appears to inhibit effective activation of the SHS muscles. Therefore, pain must be well controlled.

The initial step involves educating the patient in the role of the SHS muscles to provide dynamic hip stability, and the location and actions of these muscles. The second step involves facilitating independent contraction of these muscles. This is often best commenced in 4-point kneeling (Fig. 31.18a), where the patient is taught to activate the SHS muscles and then perform an isometric external rotation or adduction contraction against minimal resistance. The aim is to produce a low-level tonic hold of these muscles. In this position (90° hip flexion), the contribution from the larger external rotator (gluteus maximus) is reduced (see section on joint structure



(a)

Figure 31.18 Deep hip stabiliser strengthening exercises
(a) Activation of the SHS muscles in 4-point kneel with TheraBand™ resistance. The degree of difficulty can be progressed by decreasing or increasing the level of resistance, changing the speed of activation and increasing the number of repetitions. The challenge to the core can also be increased by lifting one hand off the floor in this position, and the degree of hip flexion or extension and abduction or adduction can be altered based on the needs of the athlete



(b)

Figure 31.18 (cont.) (b) Progression of activation of deep hip stabilisers into a closed chain position, ensuring adequate deep hip external rotators, gluteus maximus, lateral pelvic and core stability

and muscle function), thus enabling more specificity of activation for the SHS muscles.

Both the patient and the clinician must be confident that the deep hip stabilisers are activated and a real-time ultrasound (RTUS) machine may assist with providing feedback. Progression of the retraining includes providing different levels of resistance, number of repetitions and speed of movements. Other progressions include increasing the amount of hip flexion and decreasing the support (i.e. lifting one hand) to increase the balance demands and challenge to lumbopelvic stability.

Further progressions include activation of the deep hip stabilisers (Fig. 31.18b) in a variety of degrees of hip range of motion and in various functional positions as the activity of the athlete demands, and these can be assessed using an RTUS in these varying positions. For example, an athlete that performs regularly in positions of hip flexion such as a deep squat should ultimately perform muscle activation in this position.

Phase 2: Gluteus maximus retraining

Gluteus maximus plays an important role in generating extension, adduction and external rotation torque, and has the potential to provide hip stabilisation by resisting anterior hip force.^{33, 35} Facilitation of independent gluteus maximus contraction may be best commenced in prone (Fig. 31.19a and b), where the patient is taught to perform an isometric external rotation contraction against minimal resistance (low-level tonic hold of these muscles). As with the SHS muscles, feedback may assist in ensuring that the muscle is activated. Since the gluteus maximus is more superficial, feedback may be provided by palpation, surface electromyography biofeedback or RTUS machine.



(a)



(b)

Figure 31.19 Gluteus maximus retraining exercises. Two examples of activation of gluteus maximus, in combination with short hip external rotator (SHER) muscles in prone. RTUS assessment can also be undertaken in this position (a) Prone—knee extension (b) Knee flexion

The activation of the gluteus maximus should be undertaken in a variety of degrees of hip range of motion as the functional demands of the athlete's activity require and can be assessed using an RTUS in these varying positions. For example, hip abduction and external rotation or hip adduction and internal rotation for an athlete who performs cutting manoeuvres, or in hip flexion for an athlete that is required to perform in a deep squatting position. It should be then progressed from open chain to closed chain and then functional positions.

Phase 3: Generalised strengthening exercises

Generalised hip strengthening exercises should only be commenced when the patient and clinician are confident that the key stabilising muscles can be activated and the activation maintained. During this phase, the aim

is to restore muscle function (strength, endurance) and proprioception. This phase remains low impact (Fig. 31.20a). Exercises should initially be undertaken with specific activation of the deep stabilisers prior to completing the exercises. This ensures that the athlete has adequate control of the hip prior to placing it under load, which will assist in protecting vulnerable or damaged structures within the hip.

Generalised hip strengthening exercises should be undertaken, based on clinical assessment, for example targeting hip abductors (predominantly gluteus medius), for those with reduced hip abduction strength.

Strengthening exercises need to be targeted to the needs of the individual, progressed according to patient responses and targeted to the sporting/physical requirements. For example, an athlete who regularly jumps and lands such as a netballer or gymnast should incorporate these actions into their rehabilitation program (Fig. 31.20b).

Exercises are frequently commenced in prone (to ensure specificity and isolation of muscle activations) or in 4-point kneeling and then progressed into

functional/weight-bearing positions, bilaterally and then unilaterally (Fig. 31.20c-e). Such exercises also address the rehabilitation goals of balance and functional task performance.

3. Improve balance and proprioception

Retraining of balance can commence immediately in low-impact dynamic functional tasks and be progressed as symptoms and performance allow. It is important that during balance tasks, the athlete is reminded of correct alignment of the pelvis and femur, and to maintain activation of the SHS muscles to facilitate control.

4. Improve hip control in functional task performance

Reducing load on the anterior aspect of the hip joint may require assessment and treatment of movement patterns during functional tasks. Patients with chondrolabral pathology demonstrate reduced balance and increased femoral adduction during a single-leg squat.¹²³ This potentially places the hip into a position of impingement and loads vulnerable anterior structures.



(a)



(b)

Figure 31.20 Generalised hip strengthening exercises (a) An example of low-impact functional retraining of deep hip stabilisers, ensuring adequate activation of these muscle groups in a challenging situation without excessive impact or load through vulnerable hip structures (b) A jumping and landing task ensuring adequate activation of deep hip, lateral pelvic and core stabilisers at take-off and landing



(c)



(d)

Figure 31.20 (cont.) (c) and (d) Two examples of later-stage functional activities with concurrent SHER muscle activation, core activation, and functional balance and proprioceptive challenges

Improving the motions of the hip (i.e. reducing hip internal rotation and adduction) and trunk lateral lean may reduce load on the anterior hip. This may be addressed through hip abductor, adductor and rotator strengthening; balance, trunk strength and movement retraining programs described previously. Increased femoral adduction during single-leg tasks will result in increased impingement and may prolong the athlete's symptoms. Adequate femoral control without excessive adduction must be gained in all balance and functional tasks.

Athletes with specific strength deficits of the trunk, hip abductors and hip adductors should address these in targeted strength programs as well as in functional and balance retraining. Adequate control of the femur during single-leg tasks is essential before an athlete can return to sport (see box), especially when fatigued. This can be achieved using specific motor-control training of

functional tasks as well as training to address strength-endurance to prevent fatigue-related loss of control.

5. Improve trunk muscle strength

Trunk muscle strength is an important target for athletes with hip pain, given that it is impaired in people with hip pathology, and when reduced, increases acetabular retroversion and impingement. Principles of exercise prescription to optimise spinal stiffness and mobility are given in Chapter 11. When the patient performs trunk exercises, the clinician should ensure the exercises are pain free and the hip is not placed in positions of impingement. The patient should avoid over-using secondary trunk muscles such as iliopsoas, as this may increase impingement and hip pain. Exercises such as sit-ups, where the hip flexors are commonly used, may exacerbate these symptoms and should be avoided.



(e)

Figure 31.20 (cont.) (e) Example of a later-stage functional exercise incorporating deep hip and core stability with proprioceptive and neuromotor control retraining

6. Optimise gait biomechanics

Specific activities to normalise gait biomechanics and improve gait performance and endurance are outlined in Chapter 8 and should be progressed in a structured, graduated fashion. It should also include running, direction

change and cutting manoeuvres specific to the athlete's functional demands. If required, it should also include stairs and hill training (up and down). Generally, patients with hip pain have a reduced step length to avoid increased joint loading seen at end-range hip extension. Any modification to gait should not increase the athlete's hip pain.

7. Optimise functional task performance

Once good neuromotor control of the deep hip stabilisers and global hip muscles has been regained, functional and sports-specific activities should be assessed and then undertaken, both to retrain these movement patterns, but also ensure the athlete can cope with these activities without failing.

Any retraining of functional activities should focus on pre-activation of the deep hip stabilisers, adequate control of the lumbar spine and pelvis during the activity, and correct alignment of the femur during weight-bearing tasks (Fig. 31.21).

Retraining of hip stabilisers should be performed in the positions that place the hip at greatest risk of overload, such as direction change and pivoting, deep squatting and kicking. They should also be undertaken in a repeated fashion, again to ensure the athlete does not fail in a controlled environment.

8. Address adverse loading

The most effective way to unload and protect specific structures of the hip varies slightly for different pathologies, based on our understanding of the functional anatomy and biomechanics of the hip. When addressing the loads on structures outlined below, the principles of management of neuromotor control and remote factors should also be applied. Managing the load on the hip can be particularly difficult as the athlete may need to increase provocative loads on the hip in daily tasks. Thus it is vital that the ability to climb stairs, squat, put on shoes and perform



Figure 31.21 Retraining of functional activities—single-leg hop for distance

Criteria for returning to sport as the final stage of hip rehabilitation

The decision regarding a patient's readiness to return to sport is made using clinical judgment of the individual's functional capacity. In the absence of robust scientific evidence, the following criteria are suggested:

- performance on the one-leg hop test (or other single-leg functional tests) at least 90% of the uninjured side (if unilateral symptoms)
- performance on strength tests at least 90% of the uninjured side (if unilateral symptoms)
- performance of functional tasks and sporting activity that does not reproduce hip pain, and that the athlete has adequate control of the hip to avoid impingement, even under conditions of fatigue.

B

everyday activities such as getting in and out of a chair is undertaken in such a way that these activities do not aggravate the underlying pathology.

9. Address other remote factors that may be altering the function of the kinetic chain

As outlined previously, a number of remote factors (e.g. lumbopelvic control, ankle dorsi-flexion range) are likely to influence the rehabilitation of hip pain and pathology. Therefore, all potential contributing factors should be addressed and treated appropriately.

Surgical management of the injured hip

Hip arthroscopy is now commonly performed to manage intra-articular hip pathologies, including FAI, labral tears, chondropathy and ligamentum teres tears.¹²⁹ Hip arthroscopy has revolutionised hip surgery, since this minimally invasive procedure is associated with considerably less morbidity than open procedures. Internationally, the number of hip arthroscopy procedures now performed is growing rapidly. In the USA, the number of hip arthroscopies has increased 18-fold in the past decade⁴¹ and in Australia it has doubled in recent years. Hip arthroscopy is a technically challenging orthopaedic procedure and should only be undertaken by high-volume, fellowship-trained surgeons.

Clinically, patients presenting for hip arthroscopy surgery tend to be grouped into two categories—those diagnosed with morphological variations, with or without soft tissue injuries, requiring surgical intervention, and those not requiring bony intervention but presenting with soft tissue injuries requiring intervention.

The first group includes patients with FAI, which may be cam, pincer or mixed impingement. This group may also have coexisting labral pathology, ligamentum teres pathology or chondral lesions.

The second group includes those with soft tissue pathologies, but without morphological change requiring surgical intervention. This group may include labral pathology, ligamentum teres pathology, chondral lesions or

any combination of these. They may have coexisting issues such as dysplasia or hypermobility, which predispose them to such injuries, but do not themselves require surgical intervention. This group may also include patients with essentially normal morphology, but who undergo a massive single episode of excessive range (usually rotation), which causes trauma to the associated soft tissues.

The body of evidence examining outcomes following hip arthroscopy is growing rapidly. Systematic review evidence shows that patients without chondropathy have favourable outcomes for up to 10 years (no osteoplasty) and 3 years (with osteoplasty).¹²⁹ Patients with chondropathy have worse outcomes than those without, especially when chondral damage is severe, and risk progressing to total hip replacement within 2 years.¹⁰⁹ Prior to surgery, it is important that patients trial the conservative treatments outlined above. There are still no known published randomised controlled trials examining the effectiveness of hip arthroscopy, although these studies are underway internationally and will provide definitive evidence of the efficacy of hip arthroscopy for subgroups of patients with intra-articular hip pathology in the coming years. The majority of the literature focuses on outcomes following surgery for FAI, labral pathology, chondropathy or combined pathology, and is level III and IV evidence.

Rehabilitation following hip arthroscopy

Rehabilitation following hip arthroscopy has been described in the literature in a number of clinical commentaries and a systematic review.^{115, 117, 130, 131} Rehabilitation programs essentially follow the same conservative principles of management outlined above. The individual pathology treated during hip arthroscopy should influence the post-operative rehabilitation program to ensure it is adequately unloaded and protected whilst healing. This generally involves a period of partial weight-bearing as tolerated on crutches until a pain-free normal gait pattern is achieved.

Generally osteoplasties performed for the correction of FAI must be protected for at least 6 weeks, while

microfracture surgery performed for chondral defects may be protected through non and partial weight-bearing for at least 3 months.

Labral debridement and repairs should be protected for at least 6 weeks, ensuring the athlete avoids potential positions of impingement through activity modification and normalisation of neuromuscular control around the hip.

Injuries to the ligamentum teres should be protected for at least 6 weeks by avoiding end-range positions that place the ligament under stress, and ensuring excellent neuromotor and proprioceptive control around the hip.

During this initial protective phase, the athlete should commence active rehabilitation of the deep hip stabilisers, initially in an isolated fashion and then progressing into functional activity in a safe manner. The therapist should also address any overactivity of the secondary stabilisers such as the long adductors, the proximal gluteals, the tensor fasciae latae and the hip flexors in this period.

Once this protective phase is complete, the athlete should undertake a dynamic rehabilitation program ensuring full strength of all muscle groups around the hip, and normal function of the whole kinetic chain and sports-specific activity. A full assessment of the muscle strength and function around the hip using RTUS and hand-held dynamometry at this time can also assist in providing targeted exercise programs to address any ongoing residual deficits in strength or muscle activation. Generally, most athletes return to full sport between 3 and 5 months postoperatively following hip arthroscopy, although this will vary depending on the level and type of sport played, as well as the specific pathology and surgery performed.

LATERAL HIP PAIN

with ALISON GRIMALDI

Lateral hip pain signals very different pathologies to the medial or anterior hip region pain the chapter has focused on so far. It is a common presentation particularly among distance runners and women over the age of 40. Local soft tissue pathology may mimic or coexist with hip joint and lumbar spine pathologies so a full differential diagnosis as described earlier in this chapter will be required. Hallmarks of a local pathology include specific areas of pain and tenderness either at the greater trochanter or iliac crest.

Greater trochanteric pain

Patients with trochanteric pain complain of pain and significant tenderness over the greater trochanter, particularly lying on their side at night. The pain may radiate from the greater trochanter, most commonly down the lateral aspect of the thigh to the knee, and occasionally extending into the upper lateral leg along the line of the ITB. Tasks that involve a single-leg weight-bearing phase such as standing on one leg to dress, ascending stairs or

hills and particularly those involving higher eccentric loads or a stretch-shortening cycle, such as running, bounding or hopping, are usually reported as provocative.

Traditionally the primary local source of trochanteric pain was thought to be the trochanteric bursa; however, there are a number of soft tissue structures at the greater trochanter that have nociceptive capacity. Less common findings include thickening of one or more of the three associated bursae and the region of the ITB crossing the greater trochanter.¹³²⁻¹³⁴

PRACTICE PEARL

Imaging studies have demonstrated that soft tissue pathology in this patient population most commonly presents in the gluteus medius and/or minimus tendons.

The same pathological mechanism is likely to underlie all of these local soft tissue changes, and therefore the relative local contributions to nociception may be less pertinent in early conservative management than attending to the underlying issues. While medical comorbidities may compromise tendon function and load-bearing status,¹³⁵⁻¹³⁷ mechanical loading is a potent driver of biological processes responsible for soft tissue structure and health.^{138, 139}

Relevant anatomy

Mechanical load within the local soft tissues will be influenced by the relative anatomical relationship between these tissues and the underlying bony greater trochanter in different postures and dynamic function. As the gluteus medius and minimus tendons wrap around the greater trochanter on their course to their insertion sites, they are separated from the underlying bone by their relevant bursae (subgluteus medius and subgluteus minimus bursae) (see Fig. 31.5 for general anatomy of this region).

The trochanteric bursa, often now referred to as the subgluteus maximus bursa, sits superficial to these tendons and beneath the thick, fibrous ITB. This relationship is illustrated in Figure 31.22. Mechanical load may be applied longitudinally (tensile load) or transversely (compressive load) across the tissues. Excessive compressive load is an important aetiological factor in the development of insertional tendinopathies,^{140, 141} engendering changes within the soft tissues that may ameliorate compression but subsequently reduce tensile loading capacity.

Role of compression in pathology

The gluteal tendons and bursae may become compressed between the underlying greater trochanter and overlying ITB, while the ITB itself may also be exposed to compression against the greater trochanter. Compression increases tenocyte production of larger

PRACTICE PEARL

Compressive loading of the trochanteric soft tissues may be accumulated during static postures such as standing 'hanging on one hip' in adduction, sitting with knees crossed or together, or during dynamic functional tasks completed with excessive lateral pelvic tilt or shift.

proteoglycan molecules (aggrecan and versican)¹⁴² that bind water within the tendon resulting in a thicker tendon more resistant to compression. There may also be a shift within a compressed tendon to cartilage-like cells, which are particularly robust for compressive loading. Thickening of the associated bursae and the ITB are also likely to represent a response to excessive compression. Unfortunately this seemingly successful adaptation to compression results in reduced tensile loading capacity of a tendon due to concurrent disorganisation and enzymatic resorption of large type 1 collagen fibres.¹⁴³

Tendons that have adapted to compression may then fail at lower tensile strain rates.¹⁴¹ Consistent with this scenario, gluteal tendons in those with lateral hip pain have been shown to be either thicker than normal (likely in the earlier adaptive stage), or attenuated and thinner with partial or full thickness tearing (later stage when the compromised collagen fibres yield under tensile loads).^{132, 72}

Compressive loading of the lateral soft tissues is strongly influenced by hip joint position and particularly by hip adduction. Birnbaum and colleagues determined that, as the hip moves from neutral adduction, where the ITB exerts 4 N of compressive load on the greater trochanter and

intervening tissues, loads rise nine-fold by 10° of adduction and over 25-fold by end-range, 40° adduction.¹⁴⁴ Running with a midline or cross-midline striking pattern, on the camber of a road or the same direction around a track, may increase the risk of developing lateral hip pain,^{145, 146} perhaps due to the cumulative loading stimulus.

Iliac crest pain

Lateral hip pain may also emanate from soft tissue sources at the iliac crest, with local pain and tenderness over the region of origin of the TFL and ITB and adjacent fascia, with pain radiating most commonly over the anterolateral hip towards the greater trochanter. In such patients, thickening, increased water content and partial tearing have been noted on MRI in the TFL tendon of origin, the proximal ITB and the gluteal aponeurotic fascia that covers the gluteus medius, all of which anchor along the iliac crest.^{134, 147, 148} While an in vivo study such as Birnbaum et al.'s¹⁴⁴ has not been replicated at the iliac crest, it is reasonable to conceive that, as the hip moves into adduction, the ITB and the gluteal aponeurotic fascia will also wind firmly around the iliac crest, with the iliac tubercle providing the most prominent bony cam region. In running athletes, this condition is an overuse injury with reports of gradual onset worsening with activity over a number of months.^{134, 148} The clinician should be aware of the same issues as for greater trochanteric pain—assessing and controlling exposure to hip adduction during static postures and functional loading.

Examination of the patient with lateral hip pain

Distinct tenderness on direct palpation over the greater trochanter or iliac tubercle is a key finding that determines

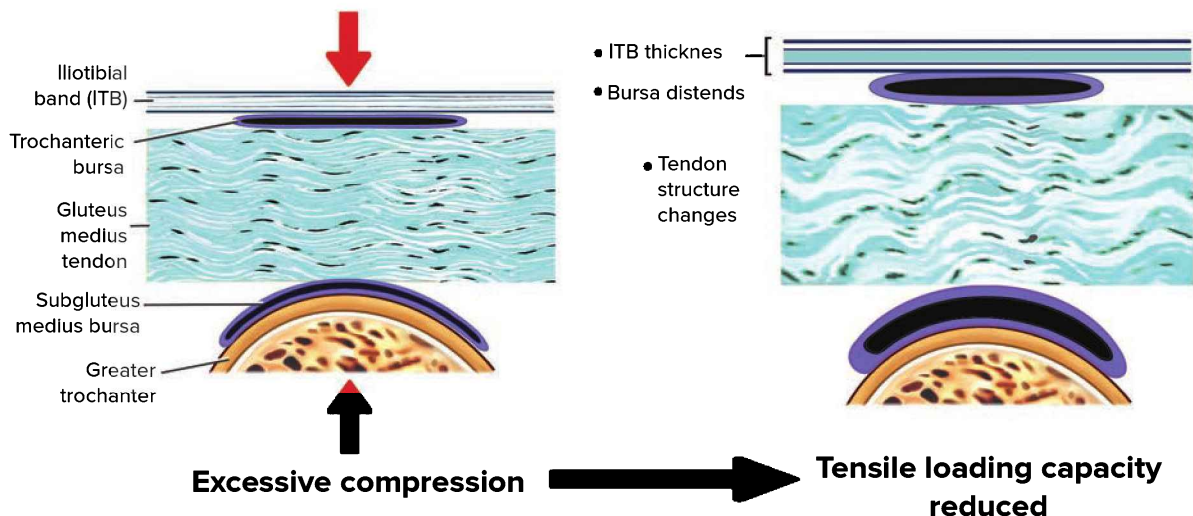


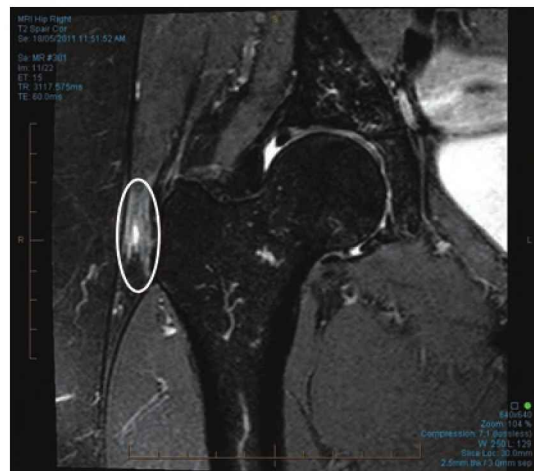
Figure 31.22 Adaptation of soft tissue of the greater trochanter to compressive loading. Structural changes include production of larger proteoglycans, a hydrophilic gel matrix, increased water content and a shift to cartilage-like cells

whether lateral hip pain is originating from a local soft tissue source. A lack of local tenderness would strongly indicate that other more remote conditions should be considered. However, tenderness alone is not sufficient for a firm diagnosis. Lumbar screening and a full hip joint assessment, as described in the preceding physical examination section, should be undertaken. Range of motion tests are usually unremarkable, helping to differentiate from primary hip joint pathology. Similarly, the FABER test is usually not significantly limited in range, but reproduction of lateral hip pain on this test, in the absence of difficulty manipulating shoes and socks (which is usually associated with limited FABER), has been shown to be useful in differentiating local greater trochanteric pain from hip OA.¹⁴⁹

Despite the anecdotal assumption that those with lateral hip pain have tightness in the lateral soft tissues of the hip and thigh, particularly the ITB, the adduction range on the Ober's or Modified Ober's test is not correlated with the presence of gluteal tendon pathology.¹⁵⁰ About 25% of participants were positive for local pain at the greater trochanter during the Ober's test.¹⁵⁰ This may reflect the potentially provocative action of compressing the soft tissues between the greater trochanter and the ITB. This adducted position will also wrap the proximal ITB and TFL tendon around the iliac tubercle and therefore may be clinically useful as part of a test battery. Diagnostic utility of the Ober's test for proximal ITB/TFL pathology is currently unknown, as no clinical tests apart from palpation have been described or analysed for such conditions.

Passive compressive tests may become more valuable with the addition of an active tensile load. Isometric abduction tests have, however, only been described to date from a hip-neutral or abducted position. While abductor muscle weakness is common in those with lateral hip pain and is important to assess, the diagnostic usefulness of such tests is limited by the fact that patients with no lateral hip pain or other conditions also commonly present with hip abductor muscle weakness.¹⁵⁰ Pain reproduction on resisted hip abduction showed some weak diagnostic properties.¹⁵⁰ Further research is required to assess whether isometric abduction performed in an adducted hip posture may be more useful in differential diagnosis of lateral hip pain.

In a meta-analysis of diagnostic accuracy of clinical hip tests, the only test to show a strong ability to influence the likelihood of a positive diagnosis of gluteal tendinopathy was the resisted external rotation de-rotation test (Fig 31.6f).⁴⁰ The test winds the ITB and fasciae latae over the greater trochanter in hip flexion and external rotation, and then superimposes an active contraction of the gluteus medius and minimus via resisted internal rotation, combining compressive and tensile loads across the greater trochanter. Adding hip adduction will further increase compressive load and therefore may be useful as a test variation.



(a)



(b)

Figure 31.23 MRI appearances of gluteus medius abnormalities (a) Coronal fat-suppressed T2 image showing tendinopathy of the gluteus medius tendon near its greater trochanteric insertion (b) Fat-suppressed T1 image showing partial tear of the gluteus medius tendon (arrows)

Functional movement tests are employed to highlight functional deficits and may provide helpful treatment direction. These tests generally do not perform well as diagnostic tests as dysfunction around the hip and pelvis is common to many lower quadrant disorders. One exception may be the 30-second sustained single-leg stance test as described by Lequesne et al.⁴⁸ When performed as a pain provocation test, rather than a traditional Trendelenburg test, reproduction of pain over the greater trochanter

within 30 seconds of standing on one leg was shown to have good sensitivity and specificity for the detection of gluteal tendinopathy.

Diagnostic ultrasound can be performed to determine if thickening is present in the bursae or ITB or to look for thickening, thinning or hypoechoic changes that are consistent with tendinopathy and tears.^{72, 132, 133} MRI is the gold standard for assessing tendinosis and tears of the gluteus medius and minimus tendons (Fig. 31.23).^{151, 152} While imaging studies are useful as an adjunct to clinical assessment where the diagnosis is unclear or the patient is failing to respond to treatment, they should not be used as a stand-alone medium for determining the source of lateral hip pain. This is due to the high prevalence of soft tissue abnormalities in this region that may occur in the absence of lateral hip pain.¹⁵²

Treatment of the patient with lateral hip pain

The principles of treating lateral hip pain are consistent with other insertional tendinopathies. First, manage the pain by controlling soft tissue loads at the greater trochanter or iliac crest. Introduce graduated strengthening of the involved musculotendinous complex, aiming to improve load tolerance and optimise abnormal movement patterns. Return to sport is closely monitored to ensure volume of tendon load is controlled, time for adaptation is allowed and recurrence is avoided.

Managing pain

Managing pain will primarily involve removing or minimising potentially provocative loads and instituting pain-relieving loading techniques. Reduction of compressive loading will require minimisation of sustained, repetitive or loaded end-range adduction. Positional habits such as sitting with knees crossed or together or standing 'hanging on one hip' in adduction should be discouraged. Sleeping postures should also be addressed, educating the patient to sleep in supine, quarter from prone or, if side-lying is unavoidable, to use pillows between the legs to reduce adduction of the uppermost side. An eggshell overlay for the lowermost side is recommended.

ITB and gluteal stretches involving hip adduction should be avoided. Provocative physical activities should be suspended or modified, such as long-distance, high-speed and hill running or, for the more severe presentation, all running may need to be temporarily suspended. Runners should stay on the flat or run in straight lines rather than on a camber or around a track; plyometric drills such as jumping, bounding and hopping particularly using stairs or boxes should be avoided.

While corticosteroid injection may reduce pain in the short term via interactions with local neuropeptides and

neurotransmitters,^{153, 154} the effect is not long lasting. It fails to address the underlying pathoetiological issues and may even hinder a tendon's capacity to respond appropriately to loading (see Chapter 17) via the down-regulation of fibroblastic production of collagen.^{155, 156}

Shock wave therapy reduces lateral hip pain.^{154, 157} However, it is recommended that shock wave therapy is reserved for those who fail a load management and exercise approach.^{158, 159}

Managing load: First line treatment

Exercise for the hip abductor musculature begins with isometric abduction in neutral or slight hip abduction to minimise compression. This may be completed in side-lying (affected side up with pillows between legs) (Fig. 31.24a), supine with a belt around the lower thighs or standing (Fig. 31.24b). The patient should *slowly* ramp up the contraction to a 25% maximum voluntary isometric contraction level initially, which allows time for a good gluteus minimus and medius recruitment, and avoids dominance of the TFL. The level of force development may be increased if well tolerated and early and dominant gluteal contraction is achieved. The primary aim at this point is pain relief, provided by isometric exercise via activation of segmental and/or extra-segmental descending pain inhibitory mechanisms,^{160, 161} and now recommended in the early management of lower limb tendon pain.^{162, 163}

The strengthening phase includes exercises that directly target the hip abductors and functional loading tasks with a focus on strict control of hip adduction through graduated levels of difficulty (Fig. 31.25). Spring-resisted sliding platforms such as Pilates reformers provide frontal plane weight-bearing resistance with applied load easily titrated and progressed.

PRACTICE PEARL

Side-lying 'clams' (hip abduction/external rotation–hip adduction/internal rotation) are generally provocative, possibly due to the compression and friction that the ITB imparts on the soft tissues as it passes over the greater trochanter; they should therefore be avoided.

Bridging (Fig. 31.26) and functional strengthening progressions (Fig. 31.27) begin with bilateral loading, progressing to offset and then single-leg tasks, allowing hand support where necessary to optimise pelvic control and minimise hip adduction and therefore compression, which is the first priority.

Athletes can then progress to more complex sports-specific tasks, as described in the management of hip joint



(a)



(b)

Figure 31.24 Examples of positions for pain-relieving isometric abduction exercises for lateral hip pain (a) Side-lying 'preparation for lifting' (b) Standing in slight abduction



Figure 31.25 Resisted hip abduction exercise (sliding) in weight-bearing



(a)



(b)

Figure 31.26 Examples of basic bridge progressions (a) Double-leg bridge—equal weight-bearing (b) Offset bridge—gluteals of 'close side' foot do most of the work



(c)

Figure 31.26 (cont.) (c) Single-leg bridge—pelvis controlled in a level position by the gluteals of the weight-bearing side throughout the hip flexion/extension movement



(b)



(a)

Figure 31.27 Examples of functional strengthening progressions (a) Double-leg squat—equal weight-bearing (b) Offset squat—gluteals of front side do most of the work (c) Single-leg squat—pelvis controlled in a level position by the gluteals of the weight-bearing side throughout the hip flexion/extension movement



(c)



(d)



(e)

Figure 31.27 (cont.) (d) Step up (e) Landing training—requires control of pelvic and femoral alignment during a dynamic landing task

pathologies earlier in this chapter. For running athletes, visual biofeedback and increases in cadence have been shown to be effective in reducing peak hip adduction.¹⁶⁴ Increasing cadence by 10% enhanced pre-activation of gluteus medius prior to stance phase and improves in lateral pelvic control during the subsequent stance phase.¹⁶⁵ As stretching into adduction is to be avoided, athletes may require massage, needling or home use of a trigger ball or vibratory hand-held massager to control

muscle soreness or tightness arising secondary to the loading program.

For those who fail conservative management, usually older non-athletic individuals, there are surgical options that include ITB release around the greater trochanter $\pm$ bursectomy and tendon repair as required.¹⁶⁶⁻¹⁶⁹ While outcomes are generally reported as good to excellent, athletes are much less likely to progress to a surgical intervention.

REFERENCES

References for this chapter can be found at www.mhhe.com/au/CSM5e